

BST Working Paper — Volume 5: The Predictions — Chapter 1: Experimental Predictions and Research Program

2026-09-21 — rewritten as the falsifier register rendered in prose
(Lyra; Casey's scope word PROSE 2026-09-19; Keeper gates). May
2026 text kept below as the record.

Volume 5, Chapter 1 — The Predictions, as the register carries them

*Rewritten 2026-09-21 (Lyra, on Casey's scope word PROSE of 2026-09-19 and K1913
Section 3; Keeper gates this chapter against the register rows). This chapter is the falsi-
fier register — notes/Elie_FALSIFIER_REGISTER_v0_2_C1_owned_and_pinned_NuFIT6_three_delta_CP_claims
08-24.md, v0.19 as of 2026-09-21 — rendered in prose. Every claim, bound, σ and
date below is taken from a register row and names the row. Where the prose and
the row disagree, the row wins and the prose is rewritten and dated. The 2026-09-19
head-note and the May 2026 text are kept below as the record. Last accuracy-synced:
2026-09-21 / register v0.19.*

The question

What has this program bet, in public, in a form that can lose — and what happened
each time we checked?

A match is not a prediction. A match is a number that came out close; the May
text below has about two hundred and fifty of them, each with a check mark. A
prediction is a bet written down before the check, with the losing condition written
beside it. This chapter is the list of bets. The register holds nine that are live, six
that were made and lost, four that ended at a wall with a named door, ten that
the volumes named as experiments and that were checked against the literature
on the day they were registered, three controls that fired the way controls should,
two claims in quarantine, and a short list of boundary markers that nobody may
promote to a bet. That is the whole inventory. Nothing else in this volume is a
prediction in the register's sense.

How to read this chapter

Four words, used exactly (Spine Lecture 10, Curriculum/Spine_DIV5_QM_GR_SM/Lecture_10_The_Method_and_Ho
derived — a mechanism proved, with its inputs named on the statement; *identified* — a formula that matches, mechanism not claimed; *open* — used from measurement; *input* — the ruler. A fifth, *partially derived*, only in explicit-split form — the derived mechanism and the identified input each named on the statement, never a blended adjective. The tier words for the 26 primary Standard-Model parameters live on the generated table, `data/bst_26_tier_table_generated.md`, and nowhere else; this chapter types none of them.

Every σ and every percent below is against a named release — PDG 2024, NuFIT 6.0, or a paper cited by DOI on the row. A falsifier that names an experiment is checked against the literature the day it is registered. The rule exists because one of ours, E4 below, had been refuted a decade before it was written.

A register that deletes its losses hides its record. The losses are here, in the same font as the wins.

To regenerate a number without trusting us: `python3 play/verify_bst.py` runs the registry's verification set; `python3 play/toy_bst_explorer.py verify T187` checks one theorem; `bash play/reproduce_A9_run1.sh` takes A9's run 1 from the catalogue download to the letter, under the frozen protocol (Keeper K1917). Every toy named below is in `play/` with its pre-registration in `notes/`.

1. Live falsifiers — register Section A

A1 — no genuine wobble in Λ , no phantom crossing. The bet: dark energy relaxes to $w = -1$ from above, monotone, never crossing (F799, T2559), and its variance is Pauli-frozen (T2571) — two facts, not one (K1749). Current bound: the phantom crossing DESI DR2 reports depends on the gravity model used to read it; a scalar-tensor reading removes it (arXiv:2511.04610), so it is not a clean detection. Kill: a confirmed, interpretation-robust crossing, or a genuine stochastic wobble. Owner: Grace.

A2 — the CKM first row is exactly unitary. Status (register v0.29, 2026-09-26): A2 fired on $K_{\mu 2}$ (Cal Sections 989/992 criterion, certified K1928 second amendment; the $|V_{ud}|$ face past 3σ ; rescues named, not credited); FLAG 2027 decides. The bet: $\lambda = 1/\sqrt{20}$ and $|V_{ud}| = \sqrt{(19/20)}$, unitary by construction — Casey's chosen observable (R71). Current: $\lambda = 0.22501 \pm 0.00068$ against 0.22361, a -2.06σ tension; the direct first-row sum, 0.9984 ± 0.0007 , is 2.3σ short of unitary, which runs AGAINST an exactly-unitary prediction (this sentence said “in BST's favour” until 2026-09-26; Cal Section 995, Keeper). Kill, as registered: a route more than 3σ from A2 that is not itself more than 3σ from another route measuring the same ratio (Cal Section 989). The $K_{\mu 2}$ route meets it (-3.25σ isospin-limit / $+3.69\sigma$ charged against a $2.3\text{--}2.7\sigma$ SM split). Owner: Keeper.

A3 — no sterile neutrino. The bet: Five-Absence — there is no ν_R ; the field mul-

tiset declares ν^c absent. Current: no confirmed sterile signal. Kill: a confirmed sterile neutrino, and Cal's I-1/I-2 mechanisms inherit the failure. Owner: Lyra.

A4 — no W_R , no gauged B–L. The bet: the surviving gauged group is exactly the Standard Model's (T2520; Cal's I-1). Current: neither observed. Kill: either discovered — the neutrino slot is then charged and the hypercharge mechanism dies, in Cal's own words (Section 735). Owners: Cal, Grace.

A5 — neutrinos are Majorana. The bet: the couplings the geometry requires are the charged Yukawas only (Cal's I-2). Current: $0\nu\beta\beta$ unobserved, parameter space open. Kill: Dirac nature established. Owner: Lyra.

A6 — the lightest neutrino is massless. The bet: $m_1 = 0$, reached by two routes — the $\nu = 0$ Wallach point (F619) and the $k = 0$ no-Higgs channel (toy 5461) — one bank until they are reconciled. Current: cosmology bounds Σm_ν from above; $m_1 = 0$ under normal ordering sets the 58.2 meV floor from below (T2563 — the same number N7 carries). Kill: a measured $m_{\text{lightest}} > 0$. Owner: Lyra, with Elie on the channel route.

A7 — Koide's $Q = 2/3$. Conditional-forced (K1749; the current statement is K1830's floor ruling); toy 5100 region-match. Current: 0.6666605 on PDG masses. Kill: an improved τ mass moving Q off $2/3$ beyond the combined errors. It sits here, among the live rows, and not among the floors of Section 3, because its kill is a measurement, not a theorem: the floor ruling fixed the claim's wording, and the row waits on data. Owner: Elie.

A8 — the order of the CKM corner. The bet is an order, not a value: the (1,3) corner is suppressed by exactly one power of t relative to the (2,3) subdiagonal, because on a three-dimensional generation space every candidate series in the mixing operator collapses by Cayley–Hamilton to three terms — derived, zero knobs (K1808, K1810; Lecture 7). Current: $|V_{ub}|/|V_{cb}| \approx 0.093$, band $[0.081, 0.108]$; the *value* is open — t is unforced, and five sealed series missed it (E6). Kill: an order is reparametrisation-invariant, so this row is wrong if the corner is ever measured at other than one power below the subdiagonal, at any scale. Owners: Grace (row); Lyra and Cal (the theorem).

A9 — one boost, not two: the radiation and matter rest frames agree. This is the volume's one falsifier with a run.

The bet. BST's Machian descent (T2564) records the exterior's state through a single channel, light; there is no separate matter channel. So the cosmic rest frame read from radiation and the one read from matter must coincide: ε , the rapidity between them, is zero. A structural consequence, not a computed magnitude (Cal's pre-registration P1–P3, filed before data; T2565; Cal Section 957). *The observable.* The Ellis–Baldwin test: the dipole of a distant-source catalogue, against the CMB dipole. *The presupposition, on the row's face.* The test reads a source-count dipole as kinematic; an intrinsic large-scale anisotropy of the sources breaks that reading and is the pre-registered escape hatch. *Prior art, consistent.* Wu & Xia 2026 (Phys. Rev. D 114, 043544), the redshift-dipole method on DESI DR1: $v = 358 (+55/-48)$

km/s against the CMB's 369.82 — one boost, in the other channel. Secrest et al. 2022 (ApJL 937, L31): the radio and quasar dipoles reject the purely kinematic null at 5.1σ joint, and the authors read the residual as intrinsic — the hatch is open in the data themselves.

The protocol. Pre-registered v1.5.2, SHA256 8013d959..., Cal Section 974, verified from the files by Keeper (K1909); the target excluded from the run's inputs. *Run 1.* Quaia ($G < 20.5$ primary, $G < 20.0$ alongside), the published selection function as fallback weights: Landing C — not decidable — on both samples. Elie's record (step vi), Keeper's instrument (vii, K1910), Cal's cold read (viii, Section 976), Keeper's certification (ix, K1911, 2026-09-17). The reason is on the run's own face: the protocol's profile-alternative trigger fired at $+7.5\sigma$ and $+5.0\sigma$ against the pre-registered bar of $2\sigma_\beta$ (protocol v1.5.2, Section 4.4), and the per-bin residual test failed in three of four non-null bins. The per-bin moments rise with redshift, which a boost does not do and clustering does not do; under the fallback weighting that is a residual selection systematic, caught three ways with the target read last.

What C means. Not a hold, not a fire. Quaia at the fallback weighting is not a test of the bet. A9 stays live; the next catalogue — DESI DR1, or Quaia re-weighted with per-bin maps on a larger machine — runs only after a fresh freeze, on Casey's word. *Kill.* A confirmed kinematic mismatch $\varepsilon \neq 0$ that survives the pre-registered hatch. The decider is the redshift dependence of the matter dipole: a kinematic dipole is redshift-independent at fixed spectral index, a clustering dipole is not. Neither a large amplitude alone nor an angle alone fires it. Owners: Grace (row); Cal (the map); Lyra (the candidates); Elie (the numbers).

2. Fired and lost — register Section E

Six bets, made under a frozen protocol, lost, and certified. Each paragraph says what fired, what died, and what survived, because a loss that takes the wrong things with it is a second loss.

E1 — the α one-shot. The bet: a bare-vertex computation would land in a frozen window, 137.000 to 137.036, tolerance 0.010 in α^{-1} units. It landed at $8\pi^3/3$ — the Hua volume of the Lie sphere, not 137 (toy 5496, half-blind; K1826, Cal-certified; the must-reject fired in the same run). Died: the winding-count story of $1/\alpha$, both legs, and the never-banked prize list that rode on it. Survived: λ_H and V_{us} on their own banks; the $(1/4\alpha)F^2$ form as structure; the coefficient, now a computed number with a geometric name. The honest state is Lecture 8's: α measured; $N_{\max} = 137$ a computed invariant; every expression for α , a_e and b_0 identified, not derived.

E2 — the thermal order of the generations. The bet, frozen as a biconditional before any number was seen: Shilov-first if and only if $E_{\text{Shilov}} > E_{\text{Cartan}} > E_{\text{bulk}}$. The computation gave $E = \nu^2$ exactly, $(0, 9/4, 25/4)$ — the order exactly inverted, by two independent computations with identical triples (Elie toy 5498 and Grace's half; K1828). Died: the thermal freeze-out order as the mechanism. Survived: the three

strata, the exact grading, and the anti-alignment — the mass ladder and the commit-energy ladder point opposite ways, and that is now the exhibit the successor lane starts from.

E3 — the commit-Boltzmann mass carrier. “Commit-Boltzmann”: a single-temperature Boltzmann weight $e^{(-E/T)}$ on the three strata’s commit energies $E = \nu^2$, read as the lepton mass ratios; “anti-alignment” (E2): the mass ladder and the commit-energy ladder run in opposite directions. The bet: a parameter-free 16/9 against PDG, with the fail line frozen at 5.00 %. Measured 1.88901 (PDG 2024) against 1.77778: a 5.90 % miss, robust across the full m_τ band (toy 5499, 8 of 8; Keeper-certified 2026-08-25). Died: the single-temperature Boltzmann reading of the anti-alignment — the simplest carrier, eliminated with a number. Survived: the anti-alignment order itself, still standing and still needing a carrier.

E4 — the sub-Tsirelson ceiling. The bet, registered in May: $S_{BST} = \sqrt{(126/16)} = 2.80624$, strictly below Tsirelson’s $2\sqrt{2}$, with the deficit in S^2 exactly $1/8$ — called at the time “the sharpest falsifier in the program”. The measurement had been made a decade earlier: Poh, Joshi, Ceré, Cabello and Kurtsiefer, PRL 115, 180408 (2015), $S = 2.82759 \pm 0.00051$, which puts BST’s ceiling 41.9σ below the data. Died: the laboratory prediction, the Bell-apparatus proposal, the Bell anchor in two papers, two Curriculum chapters as standing chapters. Survived: the integer identity as arithmetic; $\text{Tr}(B^2) = 126/16$ as an operator fact with no laboratory claim; and the Tsirelson bound itself, because BST’s Hardy-space quantum mechanics recovers standard quantum mechanics (Grace R142 G3; K1893). This row is why every named-experiment prediction is now checked the day it is registered.

E5 — the parity fold as the source of the quark 2–3 misalignment. The bet, gated by K1798: is the even/odd fold a forced orthogonal projection, and does it produce $\sin^2\theta \approx 0.00168$? The fold is forced, by three blind routes — and a projector’s spectrum is $\{0, 1\}$: the down tower sits wholly in the odd sector, the up tower wholly in the even, $\sin^2\theta \in \{0.000000, 1.000000\}$, nothing between reachable (Grace R55; K1799 certified negative). Died: the fold as the mechanism of the small angle. Survived: the fold as a genuine involution; and the worry that “smallness is the hard part” was withdrawn — a graded projector with about ten percent grading gives 2.2° from a generic vector, so the open question is order one, “why about ten percent”, not a hierarchy.

E6 — five sealed series for the magnitude of the CKM split. Five named candidate series, sealed by hash before the target band was scored (band $[0.081, 0.108]$ for the corner ratio). All five missed, all high: by at least $2.1\times$ against the band’s centre (K1808) and at least $1.85\times$ against its nearest edge (K1810); Elie verified independently (R60) that no candidate visited the required region. Died: the five series. Survived: the order of the split, A8 above; and the class — the ratio $t/(1+4t)$ is a Möbius bijection, so every ratio is realised by exactly one t , and the miss constrains the five named series only. The procedure-freeze rule came out of this row.

3. Floors — walls with named doors, register Section F

A floor is a theorem on the wall plus a door with a name. It may not be softened into “pending” and it may not be reopened by labour — only by a new pre-registration through Cal’s steering and Keeper’s gate, and Casey’s word where a lane opens.

F1 — the measure-degeneration floor. Any weight derived from the interior measure (measure_int, T754) is 0 or ∞ on a degenerate stratum, so the generation freeze-out by leadership-switch cannot be scored from it: free = 0, targets = 2, valid banked triples = 0 (two-CI: Grace’s shot, Elie toy 5502, 6 of 6). The door: Shilov surface-measure weights, finite exactly where the interior measure dies — existence-gated, banked at one of three addresses. Quotations of this wall must carry the measure_int subscript.

F2 — the thermostat closes as a structure; the exponent is fixed, and at that value the form is de Sitter’s. $\Lambda/\Lambda_P = (t_K/t_P)^{2p/(2-p)}$, the response coefficient entering legitimately at a pre-committed fork; the mismatch power p was the one free thing. It is fixed: $p = 1$ by the Peirce weight — the residual scales under the Jordan quadratic representation with weights 2, 1, 0 on the time line, the spatial three-space and the dropped direction, and the form’s domain $(0, 2)$ contains exactly the middle one (Lyra, 2026-09-21; K1919). At $p = 1$ the exponent is 2, and with the tick defined as the time to begin a commitment — Landauer’s bit at the horizon temperature — the form reads $\Lambda/\Lambda_P = (H t_P)^2$, which is de Sitter’s $\Lambda \propto H^2$: the Friedmann relation $\Lambda = 3H^2\Omega_\Lambda$ in commitment language, a consistency of the form with general relativity, not a value of Λ (K1919 Sections 6–7). The value limb is closed: it would require deriving H from the five integers, and no tick in the corpus does; H is measured because it changes. T2405’s α^{36} tick was fitted to a number that was never a time and is re-labelled (K1919 Section 1). The open object is the ledger’s filling law — how the committed count’s fraction of the horizon count grows with the scale factor, equivalently δ in $S \propto A^\delta$; a constant fraction gives $w = 0$ and is dead on sight, $\delta = 1$ exactly is Λ CDM. The conditional cap (K1057 lineage) stands on the form.

F3 — the write channel. Exactly one typed zero-choice assembly survives, and it coincides with the compression onto the known closed three-dimensional algebra: the spin-3 scalar is zero structurally; four checks, three could fail, none did. The door: new dynamics for the five quadrupole directions. Derived-conditional on the Internal-A floor.

F4 — the absence floor: no group anywhere in the geometry acts on the mediator space V_{12} as colour (Cal Section 946; K1890; Elie toy 5751, hashed before the run). The V_{12} -preserving subgroup restricts to $\mathbb{C}^*\cdot O(3, \mathbb{C})$, whose maximal compact has Lie dimension 4; $SU(3)$ has dimension 8, and a compact group does not embed in one of smaller dimension. So “multiplicity 3 = colour” is a numerical fit with one measured input, and the bridge that would make it a mechanism is proved absent within the geometry. The door: a group from outside the geometry acting on V_{12} as the complex fundamental — the shape of any mechanism is now known. Lecture 2 calls this a floor and cites this row.

4. Named experiments from the volumes — register Section N

Ten predictions that the Guide and Curriculum stated with an experiment’s name attached and that had no register row. Grace registered eight on 2026-09-19 and two more on 2026-09-21, and checked each against the literature the same day. None fired. None is derived. The status words are the register’s: *live*, *not tested*; *not testable as stated*; *constrained*.

N1 — Casimir $F_{\text{asym}}/F_{\text{sym}} = 7$. Not testable as stated, with a provenance defect: the two boundary conditions are never named, and the toy every site cites (1567) contains no Casimir content. The D-tier label is retired (K1913 Section 1). The only sourced neighbour is Boyer 1974’s $7/8$ — a repulsive conductor–permeable-plate force, which predates BST by fifty-two years and is $7/8$, not 7.

N2 — Cs-137 decay in a 100 nm cavity. Live, not tested. The registered number is $\tau_{\text{inside}}/\tau_{\text{outside}} = 1 + 3/1507$, a 0.199 % shift (T2362; toy 3028) — not “parts per million”, which the volume said and no source supports. Every published environmental bound on the decay constant sits at 10^{-4} to 10^{-6} (Bellotti 2012; DDEP 2024), and none is a cavity test. Kill, as T2362 states it: a measured shift below 10^{-4} at $L = 100$ nm.

N3 — an optical-clock signature at 10^{-19} . Not testable as stated. Clocks at that level exist (Aeppli et al. 2024, 8.1×10^{-19}), but the row’s only observable is a 5.3×10^{-5} relative correction to an Allan deviation, and an Allan deviation’s own uncertainty on any finite record is per cent. A re-statement is owed: what fractional-frequency signature, at what averaging time, with what coefficient.

N4 — a five-loop deviation at $\alpha^6 \approx 1.5 \times 10^{-13}$ in a Penning trap. Constrained, not decidable. The precision is already reached (Fan et al. 2023, 1.3×10^{-13}), but the two best α inputs disagree by 5.5σ , which blocks any test below about 10^{-12} ; and the six-loop QED term is 10^{-16} , so the row must say whether it means a shift of that size or a term of that order.

N5 — eigentone signatures in atomic-clock infrastructure. Not testable as stated: the lowest catalogued eigentone is a 511 keV γ , five to eleven orders above any clock, and “enhanced vacuum activity” is not defined as a measurable. The volume’s SP-30-1/SP-30-2 labels collide with the registry’s.

N6 — the quantum-eraser revival deficit, $1/137$. Live, not tested. The bet: revived visibility is $1 - 1/137$ of untagged visibility, one-sided. The best published eraser (Ma et al. 2013) carries a 1.8 % error bar against a 0.73 % effect. Kill: the ratio equal to 1 at better than 0.3 %. The corpus also carries a memo saying the framework predicts nothing different from standard QM here; that inconsistency is flagged to Keeper and is not resolved by this chapter.

N7 — normal neutrino mass ordering. Live, not decided. NuFIT 6.0 prefers normal ordering only with Super-Kamiokande atmospheric data, $\Delta\chi^2 = 6.1$; DESI DR2 bounds $\Sigma m_\nu < 64.2$ meV with a preference for normal, 6 meV above the 58.2 meV floor that A6 carries; JUNO’s first 59 days do not decide. Kill: inverted ordering at

5σ by oscillations.

N8 — flerovium-298 as an island of stability. Not testable — no route reaches $N = 184$; the heaviest confirmed isotope is ^{289}Fl at 1.9 s — and not distinct from the standard shell-model expectation. No theorem in the registry states a Fl-298 half-life; the volume’s “hundreds to thousands of years” lies inside a literature spread of ten orders of magnitude.

N9 — a circular-polarisation floor $\text{CP} = \alpha = 0.73\%$ at black-hole horizons. Not testable as stated (the floor); undecidable (mass- and frequency-independence). The claim comes from a March paper draft with no toy, no theorem row and no audit of its fit, whose Faraday escape term carries three free parameters against eight points. The primaries: Sgr A* daily CP runs from -0.41% to -1.0% with 50 % variability, below ALMA’s guaranteed CP accuracy (Albentosa-Ruiz et al. 2026); at 4.8 GHz, $-0.36 \pm 0.05\%$ (Bower et al. 1999) — not the corpus table’s 0.31 ± 0.13 ; M87* resolved $|\nu| < 3.7\%$, image-integrated $< 1\%$ (EHT M87* Paper IX, 2023). Values below the “floor” are published on both objects, and the model absorbs any of them by signed cancellation, so the floor cannot fire in either direction. Nothing at the horizon has been measured: the EHT ring is emission at several GM/c^2 , where the corpus’s own radial formula gives 0.24–0.49 %. Rulings: Keeper K1916 retired the March fit as evidence (three free Faraday parameters on eight points, $\chi^2_{\text{red}} = 0.22$; one data entry disagreeing with its own citation in sign); Lyra and Cal (Section 979; 2026-09-21): the claim is an emission-radius statement, α a supremum, and the Faraday-free observable is the frequency-independent asymptote of the single-epoch ring CP — carried as N9a’s clause (register v0.21). Kill-condition, if the claim is ever re-stated: a ring CP at fixed epoch, with the Faraday term independently constrained from the linear polarisation, sitting below the geometric value at the emission radius.

N10 — an EHT shadow deviation from Kerr. A sentence, not a prediction. The volume’s word “shape” has no source. The only number in the corpus, $(27/2)(1 + \text{rank}/N_{\text{max}}) = 13.70$, has no unit, no derivation, no toy and no theorem row. Read literally as a diameter in GM/c^2 it is excluded at 4σ by the Sgr A* 2017 measurement (EHT Sgr A* Paper VI, 2022) and sits 1.9σ high on M87* (Paper VI, 2019); read as a fractional deviation of $+1.5\%$ on an unstated base it is undecidable at today’s precision. Its author is Lyra. The number is withdrawn on this date at its three sites (notes/BST_What_Gets_Wrong.md, notes/Paper83_Draft.md, Vol 1 Ch01) with a dated bracket; nothing replaces it.

5. Fired and survived — register Section B

Three controls, kept because a register that deletes its passes hides its record: the $\nu = 1/2$ reduction point, attached as a kill-condition to a map that later died, was confirmed (B1); a 1/30 nudge to the hypercharge assignment broke all four anomaly sums exactly as required (B2, toy 5483); a fake flipped sector failed the grid-parity must-reject as it should (B3, toy 5484). These are the instrument’s proof that it can fail.

6. Quarantine and markers — register Sections C and D

C1 — the PMNS phase δ_{CP} . Three live corpus claims on one observable: 309° (March), 77° (T2018-era), and $|\sin \delta| = 2/7$, about 197° (K1024/K1029, July). Pinned against NuFIT 6.0 normal ordering, $\delta_{\text{CP}} = 212^\circ (+26/-41)$: the 197° claim is compatible at about 0.4σ ; the other two sit at 3.3σ and 3.7σ . The ruling on which claim is live is owed before the row can move to Section A. C2 — $y_t = 1$ exactly. Observed 0.9924, a 0.78 % miss on a row that says exact; parked with a named blocker until “exact” versus “near-saturation” is adjudicated.

Markers. $H^2(\Sigma; \mathbb{Z}_2) = 0$ refutes in advance any derivation of the Pin type from the boundary’s geometry — a permanent boundary, not a prediction. The free-fermion chamber formula does not produce the Standard Model’s chirality sorting (toy 5479). The $\varepsilon = 0$ frame-agreement marker of 2026-09-13 was promoted to A9 the next day once the observable and the map were named; the marker is kept as the trail.

7. What is not in this chapter, and where it lives

- The 26 primary Standard-Model parameters. On the generated table, `data/bst_26_tier_table_generated.md`, each with its tier word, its basis rows and its last re-tier date. This chapter copies none of them.
- The Millennium problems. Attempts, per problem, on the referee-consensus scale (K940); the per-problem words are in the 2026-08-16 honest ledger and on Vol 4’s head-notes (Cal Section 977). Not predictions.
- The May check-mark tables below. Matches. Where the registry gives one of them a tier word, that word is the registry’s, not this chapter’s. They are not carried forward as predictions.
- The May “null predictions” — no monopoles, no supersymmetry, no dark-matter particles. No register row today. The May argument for each runs on the 2022 substrate picture, which the register does not carry. If they are to be bets they enter through Section N, with the literature check on the day.
- The May tables’ “CP floor at BH horizon — data exists” (May 43.5, 43.7) and Vol 1’s “EHT shadow shape”. Registered 2026-09-21 as N9 and N10 (Section 4 above): the first not testable as stated, the second a sentence whose number its author has withdrawn. Neither is a bet.

Tier line for this chapter

- Derived: the *order* of the CKM corner (A8); $m_1 = 0$ (A6, two routes, one bank); w relaxing to -1 from above with no crossing (A1, F799/T2559); the absence floor’s wall (F4); the α coefficient $8\pi^3/3$ as a computed number (E1).
- Identified: every expression for α , a_e and b_0 (Lecture 8); Koide’s $Q = 2/3$ as conditional-forced (A7); the ten Section N claims at the registry’s own words, none derived.
- Open / input: the CKM corner’s value (A8, E6); the ledger’s filling law (F2’s door; p itself is fixed at 1); δ_{CP} ’s live claim (C1).

- Posit: nothing in this chapter.
- Fired and lost, certified: E1–E6.
- Retired with this rewrite: the EHT shadow number $(27/2)(1 + \text{rank}/N_{\text{max}})$, by its author (N10); “no free parameters to adjust”; α as the Wyler volume reading; the $\mathbb{Z}$ column as a tier column; “Proved”, “Exact” and “Established” as tier words; the Σm_ν and $|m_\beta\beta|$ bands and the $0\nu\beta\beta$ “clean binary test” as predictions; the sub-Tsirelson ceiling; the timeline and summary tables as the register.
- Not claimed: any count of predictions; that any Millennium problem is proved; that the May null predictions are register rows.

What would make this chapter wrong: a register row that changes one of the sentences above — at which point the sentence is rewritten and dated, not the chapter. A Section-A row that fires goes to Section E with its ceremony, and this chapter records it in the same font.

Last accuracy-synced: 2026-09-22 / register v0.21 / K1916 reader’s notes applied / K1919 Sections 6–7 (F2).

Head-note of 2026-09-19 (kept as the record; folded into the rewrite above)

Head-note added 2026-09-19 (Lyra, on Keeper K1912 Section 2 and Section 4(b); Keeper PASS, K1913 Section 3). The chapter body below is the May 2026 text, unedited in this pass. It was written before the falsifier register, the tier words of Spine Lecture 10, and the generated 26-parameter table existed, and it does not know that a falsifier has run. Read it with this head in hand. Casey’s scope word (head-note only, or Volume 5 rewritten as the register rendered in prose, K1912 4(c)) is pending; this head stands either way.

What the body claims that the register no longer carries

- “There are no free parameters to adjust” (43.1) and every “parameter-free” heading — retired (K1912). The program’s honest sentence is Lecture 10’s: one named input, and a tier word on every result (Curriculum/Spine_DIV5_QM_GR_SM/Lecture_10_The_Method_and_How_to_Kill_It.md). The “three structural inputs” paragraph ($S^2 \times S^1$ substrate, S^1 fiber, contact graph) is the 2022 picture; the object is D_{IV}^5 , and the register’s word for its selection is Lecture 2’s — one obstruction, ruled (Cal Section 950).
- $\alpha^{-1} = 137.036$ as the Wyler volume reading (43.2 rows 1–2; “Wyler constant origin ... Derived”). FIRED AND LOST: the forced vertex computes to $8\pi^3/3$ under a pre-registered protocol (register row E1; K1826; toy 5496). The current tier is Lecture 8’s: α measured; $N_{\text{max}} = 137$ a computed invariant; every expression for α , a_e and b_0 identified, not derived.

- The `?` column with no tier column (about 250 rows in 43.2). A `?` is a match, not a tier. For the 26 Standard-Model primaries the words are on `data/bst_26_tier_table_generated.md` and nowhere else; this chapter types none of them. Cases K1912 checked by hand: $\lambda_H = 1/\sqrt{60}$ (a form the register does not carry — no row); a_e “QED exact” (identified, not derived — Lecture 8, K1872); Λ “`?` derived” (the generated table’s word governs; no value is claimed here); nuclear magic numbers “Exact” and proton stability “Proved” in 43.8 — identifications, and “Proved” as a tier word is retired for them by K940’s rule.
- Σm_ν and $|m_{\beta\beta}| \in [1.4, 3.7]$ meV as predictions (43.4, 43.7). Retired as predictions with Vol 3 Ch05’s head: $m_1 = 0$ is derived (register A6); m_{ν_2}, m_{ν_3} are open (generated-table rows 25–26). Neutrino normal ordering (43.7) has no register row.
- Named-experiment predictions with no register row. K1912 lists eight across the Guide and Curriculum; in this chapter, the Casimir modification $\Delta F/F \sim 10^{-7}$ (44.4) and the BH-ringdown “Casimir fine structure” (43.4) have no row either. The standing rule: a falsifier that names an experiment is checked against the literature the day it is registered. Grace’s rows are owed, Casimir $F_{\text{asym}}/F_{\text{sym}} = g$ first. SP-30-5, the sub-Tsirelson ceiling, FIRED against Poh et al. 2015 at 42σ , a decade before it was written (register E4; K1893). The Tsirelson row in 43.2 is the identification that survived; the ceiling did not.
- “Dark energy $w \neq -1$ — Prediction confirmed in direction” (43.7 Summary row; 43.5 row 6) — withdrawn as a confirmation (Keeper, 2026-09-25, on K1921). The register’s direction is A1’s: w relaxes to -1 from above, monotone, no crossing (F799, T2559 — in CPL, $w_a > 0$). DESI DR2’s best fit has $w_a < 0$ and a crossing: that is a tension with the register’s row, not a confirmation of it. The magnitude is not “open” in the May sense either — it waits on the ledger’s filling law (F2’s door), whose direction is to be printed blind.
- The “Falsifiability by Timeline” and “Summary” tables (43.5, 43.7). Not the register. The register is `notes/Elie_FALSIFIER_REGISTER_v0_2_C1_owned_and_pinned_NuFIT6_three_de` 08–24.md (v0.17 as of 2026-09-18): Section A, nine live falsifiers A1–A9; Section E, six fired-and-lost E1–E6 with dates; Section F, four floors F1–F4; Section C quarantine; Section D structural markers.

What the body does not know

A9 has run. The frame-agreement falsifier (radiation rest frame against matter rest frame: two boosts or one), pre-registered as protocol v1.5.2 (SHA256 8013d959..., Cal Section 974, K1909), run 1 on Quaia ($G < 20.5$ primary, $G < 20.0$ alongside): LANDING C, not decidable, on both samples, by the protocol’s own Section 4.4 profile-alternative trigger and the per-bin residuals, target read last (Cal Section 976 step viii; certified K1911, 2026-09-17). A9 stays LIVE; the next catalogue is Casey’s decision. That is this volume’s one falsifier with a run, and it is the sentence the chapter should open with.

Tier line for this chapter

- Derived / identified / open / input: by reference to the generated table and the register — never by the $\mathbb{Q}$ column below.
- Retired with this head (K1912): “no free parameters”; α from the Wyler volume; $\mathbb{Q}$ as a tier; the $0\nu\beta\beta$ band and Σm_ν , as predictions; “Proved”/“Exact” in 43.8 as tier words; the sub-Tsirelson ceiling.
- Not claimed: any count of predictions; that any Millennium problem is proved (K940: attempts, per problem); that the near-term tests in 43.7 are register rows.

What would make this head wrong: a register row that changes one of the words above — then the sentence is rewritten and dated, not the head.

Last accuracy-synced: 2026-09-19 / K1912.

May 2026 record

Below: the May 2026 text of Sections 43–44, unedited on 2026-09-21, headings renamed “May 43.x / May 44.x”. Its $\mathbb{Q}$ column is a match column, not a tier column; its “Established”, “Proved” and “Exact” are the May program’s own words and carry no tier. Read with the prose above in hand.

May 43. Experimental Predictions and Falsifiability

May 43.1 The Economy of the Framework

BST has three structural inputs: a 2D substrate with S^2 topology, an S^1 communication fiber, and the requirement that the resulting contact graph be self-consistent. From these inputs the framework derives the bounded symmetric domain D_{IV}^5 as the configuration space, the channel capacity 137, and Haldane exclusion statistics with parameter $g = 1/137$. Everything else follows. There are no free parameters to adjust, no compactification geometries to choose, no landscape of vacua to navigate. The predictions either match observation or the framework is wrong. Few moving parts means few places to hide.

May 43.2 Parameter-Free Predictions (Established)

Read before the table: the $\mathbb{Q}$ column is a match column, not a tier column, and “Established” is the May program’s word. A row here can be -2σ on the register (A2 above) and $\mathbb{Q}$ here; the register’s sentence governs. Tier words for the 26 primaries are on `data/bst_26_tier_table_generated.md` only.

Prediction	BST Value	Observed	Status
Fine structure constant α^{-1}	137.036082 (Wyler, D_{IV}^5 volume)	137.035999 (CODATA)	✓ 0.0001%
α^{-1} independent derivation	137.035 (cost function + Bergman mixing, D_{IV}^5 mode density)	137.036 (Wyler)	✓ 5 ppm
Cosmological constant Λ	$F_{\text{BST}} \times \alpha^{56} \times e^{-2} = 2.8994 \times 10^{-122}$	2.90×10^{-122} Pl	✓ 0.02%
Vacuum free energy F_{BST}	$\ln(N_{\text{max}} + 1)/(2n_C^2) = \ln(138)/50 = 0.098545$	0.09855 (partition fn)	✓ exact
Committed contact scale d_0/ℓ_{Pl}	$\alpha^{2(n_C+2)} \times e^{-1/2} = \alpha^{14} \times e^{-1/2} = 7.365 \times 10^{-31}$	7.37×10^{-31} (from observed Λ)	✓ 0.005%
Phase transition temperature T_c	$N_{\text{max}} \times 20/21 = 130.476$ BST units	130.5 BST units (partition fn)	✓ 0.018%
GUT coupling N_{GUT}	$4\pi^2 \approx 39.48$	~ 40 (1.3%)	✓
Weinberg angle $\sin^2 \theta_W$	$N_c/(N_c + 2n_C) = 3/13 = 0.23077$	0.23122 (MS-bar)	✓ 0.2%
W boson mass m_W	$m_Z \sqrt{10/13} = 79.977$ GeV	80.377 GeV	✓ 0.5%
Strong coupling $\alpha_s(m_p)$	$(n_C + 2)/(4n_C) = 7/20 = 0.350$	~ 0.35	✓ $\sim 0\%$
Strong coupling $\alpha_s(m_Z)$	Geometric β -function, $c_1 = 3/5$	0.1175	✓ 0.34%
Baryon asymmetry η_b	$(3/14)\alpha^4 = N_c/(2g) \times \alpha^4$ (T929)	$(6.104 \pm 0.058) \times 10^{-10}$	✓ 0.45%
Hubble constant H_0	67.29 km/s/Mpc (Route C: full CAMB, Toy 677)	67.36 ± 0.54 (Planck)	✓ 0.1%
Neutrino mass m_{ν_3}	$(10/3)\alpha^2 m_e^2/m_p = 0.0494$ eV	≈ 0.050 eV	✓ 1.8%
Neutrino mass m_{ν_2}	$(7/12)\alpha^2 m_e^2/m_p = 0.00865$ eV	≈ 0.00868 eV	✓ 0.35%
Neutrino mass m_{ν_1}	0 (exactly, Z_3 Goldstone)	< 0.009 eV	✓ prediction
Solar splitting Δm_{21}^2	$m_{\nu_2}^2 = 7.48 \times 10^{-5}$ eV ²	$(7.53 \pm 0.18) \times 10^{-5}$ eV ²	✓ 0.7%
Mass ratio m_{ν_3}/m_{ν_2}	$40/7 = 5.714$	≈ 5.79	✓ 1.4%
PMNS $\sin^2 \theta_{12}$	$(3/10)(44/45) = 0.2933$ (T1446)	0.2935 ± 0.012	✓ 0.06%

Prediction	BST Value	Observed	Status
PMNS $\sin^2 \theta_{23}$	$(4/7)(44/45) = 0.5587$ (T1446)	0.561 ± 0.018	✓ 0.40%
PMNS $\sin^2 \theta_{13}$	$1/(N_c^2 n_C) = 1/45 = 0.02222$	0.02203 ± 0.00056	✓ 0.9%
CKM Cabibbo angle $\sin \theta_C$	$1/\sqrt{20} = 0.22361$ (T2530, DERIVED blind 2026-07-29; the $2/\sqrt{79}/T1444$ form is retired) — exactly first-row unitary	0.22501 ± 0.00068	✓ 0.004%
CKM CP phase γ	$\arctan(\sqrt{n_C}) = \arctan(\sqrt{5}) = 65.91^\circ$	$65.5^\circ \pm 2.5^\circ$	✓ 0.6%
Wolfenstein $\bar{\rho}$	$1/(2\sqrt{2n_C}) = 1/(2\sqrt{10}) = 0.158$	0.159 ± 0.010	✓ 0.6%
Wolfenstein $\bar{\eta}$	$1/(2\sqrt{2}) = 0.354$	0.349 ± 0.010	✓ 1.3%
Jarlskog invariant J_{CKM}	$\sqrt{2}/50000 = 2.83 \times 10^{-5}; 50000 = n_C^5 \times (2^{\text{rank}})^2$	$(2.77 \pm 0.11) \times 10^{-5}$	✓ 2.1%
CKM $\ V_{ub}\ $	$A\lambda^3/\sqrt{C_2} = 1/(50\sqrt{30}) = 0.003651$	0.003660 ± 0.000110	✓ 0.25%
Higgs quartic λ_H	$\sqrt{2/n_C!} = 1/\sqrt{60} = 0.12910$	0.12938 (from m_H)	✓ 0.22%
Higgs mass (Route A)	$v\sqrt{2\sqrt{2/5!}} = 125.11$ GeV	125.25 ± 0.17 GeV	✓ 0.11%
Higgs mass (Route B)	$(\pi/2)(1-\alpha)m_W = 125.33$ GeV	125.25 ± 0.17 GeV	✓ 0.07%
Fermi scale v (Higgs vev)	$m_p^2/(g \cdot m_e) = 36\pi^{10} m_e/7 = 246.12$ GeV, $g=7=p+q$ (not the genus)	246.22 GeV	✓ 0.046%
W boson mass m_W (Route B)	$n_C m_p/(8\alpha) = 80.361$ GeV	80.377 GeV	✓ 0.02%
Top quark mass m_t	$(1-\alpha)v/\sqrt{2} = 172.75$ GeV	172.69 ± 0.30 GeV	✓ 0.037%
Number of colors N_c	3 (from Z_3 center)	3	✓
Baryon = 3 quarks	Required by Z_3 closure	Observed	✓
Proton charge radius r_p	$\dim_{\mathbb{R}}(\mathbb{CP}^2)/m_p = 4\hbar/(m_p c) = 0.8412$ fm	0.84075 ± 0.00064 fm (CODATA)	✓ 0.058%
Proton/electron mass ratio	$(n_C + 1)\pi^{n_C} = 6\pi^5 = 1836.118$	1836.153	✓ 0.002%

Prediction	BST Value	Observed	Status
Muon/electron mass ratio	$(24/\pi^2)^6 =$ $[K_3(0, 0)/K_1(0, 0)]^{\dim_{\mathbb{R}}(D_{IV}^3)} =$ 206.761	206.768	✓ 0.003%
Tau/electron mass ratio	$(24/\pi^2)^6 \times (7/3)^{10/3} =$ 3483.8	3477.2	✓ 0.19%
Quark ratio m_s/m_d	$4n_C = 20$	$20.0 \pm \sim 5\%$	✓ $\sim 0\%$
Quark ratio m_t/m_c	$N_{\max} - 1 = 136$	$135.98 \pm \sim$ 1%	✓ 0.017%
Quark ratio m_b/m_τ	$g/N_c = 7/3 = 2.333$ ($g = p + q$, not the genus)	$2.352 \pm \sim$ 1%	✓ 0.81%
Quark ratio m_b/m_c	$\dim_{\mathbb{R}}/N_c = 10/3 =$ 3.333	$3.291 \pm \sim$ 2%	✓ 1.3%
Quark ratio m_c/m_s	$N_{\max}/\dim_{\mathbb{R}} = 137/10 =$ 13.7	$13.6 \pm \sim 2\%$	✓ 0.75%
Up quark mass m_u	$N_c \sqrt{2} m_e = 3\sqrt{2} m_e =$ 2.169 MeV	$2.16^{+0.49}_{-0.26}$ MeV	✓ 0.4%
Down quark mass m_d	$(13/6) \times 3\sqrt{2} m_e = 4.694$ MeV	$4.67^{+0.48}_{-0.17}$ MeV	✓ 0.4%
Quark ratio m_d/m_u	$(N_c + 2n_C)/(n_C + 1) =$ $13/6 = 2.167$	$2.117 \pm$ 0.038 (FLAG)	✓ 1.3σ
Neutron-proton Δm	$91/36 \times m_e = 1.292$ MeV ($91 = 7 \times 13$, $36 = 6^2$)	1.2934 MeV	✓ 0.13%
Chiral condensate χ	$\sqrt{n_C(n_C + 1)} = \sqrt{30} =$ 5.477	5.452 (from m_π)	✓ 0.46%
Pion mass m_π	$25.6 \times \sqrt{30} = 140.2$ MeV	139.57 MeV	✓ 0.46%
Pion decay constant f_π	$(m_p/10)(1 -$ $(\text{rank}/N_c)(m_\pi/m_p)^2) =$ 92.4 MeV	92.1 MeV (FLAG)	✓ 0.41%
Newton's G (gravitational constant)	$G = \hbar c (6\pi^5)^2 \alpha^{24}/m_e^2;$ $12 = 2C_2$, $C_2=6$ Bergman round trips	$6.679 \times$ 10^{-11}	✓ 0.07%
Hierarchy $m_e/\sqrt{m_p m_{\text{Pl}}}$	$\alpha^{n_C+1} = \alpha^6$	$1.5098 \times$ 10^{-13}	✓ 0.017%
CMB spectral index n_s	$1 - n_C/N_{\max} =$ $1 - 5/137 = 0.96350$	$0.9649 \pm$ 0.0042 (Planck)	✓ 0.3σ
Scalar amplitude A_s	$(3/4)\alpha^4 =$ $N_c/(2^{\text{rank}} N_{\max}^4) =$ 2.127×10^{-9}	$(2.1005 \pm$ $0.0286) \times$ 10^{-9} (Planck)	✓ 0.92σ

Prediction	BST Value	Observed	Status
Tensor-to-scalar ratio r	≈ 0 ($T_c \ll m_{\text{Pl}}$)	< 0.036 (BICEP)	✓ consistent
Neutron lifetime τ_n	Fermi theory with BST inputs (G_F , $ V_{ud} ^2$, Δm , $g_A = 4/\pi$, full radiative corrections) = 878.1 s	878.4 ± 0.5 s (bottle)	✓ 0.03%
Axial coupling g_A	$4/\pi = 1.2732$ (candidate)	1.2762 ± 0.0005	✓ 0.23%
Lithium-7 ${}^7\text{Li}/\text{H}$	$\Delta g = 7$ ($p + q = 7$, not the genus) DOF at $T_c = 0.487$ MeV; reduces ${}^7\text{Li}$ by $2.73\times$	$\sim 1.7 \times 10^{-10}$ vs obs	✓ 7%
Strong CP: θ_{QCD}	$\theta = 0$ (exact); D_{IV}^5 contractible $\Rightarrow c_2 = 0 \Rightarrow \theta$ -term vanishes	$ \theta < 10^{-10}$	✓ exact
Proton spin $\Delta\Sigma$	$N_c/(2n_C) = 3/10 = 0.30$	0.30 ± 0.06 (COM-PASS/HERMES)	✓ 0%
Fermion generations N_{gen}	$ (\mathbb{CP}^2)^{Z_3} = N_c = 3$ (Lefschetz)	3 (LEP Z -width)	✓ exact
Deuteron binding B_d	$(50/49)\alpha m_p/\pi = 2.224$ MeV (T927)	2.2246 ± 0.0001 MeV	✓ 0.03%
Electron $g-2$	$a_e = \alpha/(2\pi)$ (coupling per S^1 circumference); BST $\equiv$ QED	0.00115965218 (12 digits)	✓ QED exact
Dirac large number N_D	$\alpha^{-23}/(6\pi^5)^3 = \alpha^{1-4C_2}/(C_2\pi^{n_C})^3$; exponent $23 = 4C_2 - 1$	2.274×10^{39}	✓ 0.18%
Baryon resonance $N(2190)$	$C_2(\pi_7) \times \pi^5 m_e = 14\pi^5 m_e = 2189$ MeV ($k=7$, spin $7/2^-$)	2100–2200 MeV (PDG 4★)	✓ within range
Baryon resonance ($k=8$)	$C_2(\pi_8) \times \pi^5 m_e = 24\pi^5 m_e = 3753$ MeV	undiscovered?	prediction
ρ meson mass m_ρ	$n_C\pi^{n_C} m_e = 5\pi^5 m_e = 781.9$ MeV; $m_\rho/m_p = n_C/C_2 = 5/6$	775.26 ± 0.25 MeV (PDG)	✓ 0.86%
ω meson mass m_ω	$n_C\pi^{n_C} m_e = 5\pi^5 m_e = 781.9$ MeV (isoscalar partner of ρ)	782.66 ± 0.13 MeV (PDG)	✓ 0.10%
Meson/baryon ratio m_ρ/m_p	$n_C/(n_C + 1) = 5/6 = 0.8333$; meson needs n_C slots, baryon C_2	0.8263	✓ 0.86%

Prediction	BST Value	Observed	Status
Pion charge radius r_π	$\sqrt{6}/(n_C \pi^{n_C} m_e) _{\text{NLO}} =$ 0.656 fm (NLO, VMD+ChPT)	$0.659 \pm$ 0.004 fm	✓ 0.5%
ϕ meson mass m_ϕ	$(N_c + 2n_C)/2 \times \pi^{n_C} m_e =$ $(13/2)\pi^5 m_e = 1016.4$ MeV	$1019.461 \pm$ 0.016 MeV (PDG)	✓ 0.30%
K^* meson mass m_{K^*}	$\sqrt{n_C(N_c + 2n_C)}/2 \pi^{n_C} m_e =$ $\sqrt{65/2} \pi^5 m_e = 891.5$ MeV (geometric mean)	$891.67 \pm$ 0.26 MeV (PDG)	✓ 0.02%
Kaon charge radius r_{K^+}	$\sqrt{6}/m_{K^*}(\text{BST}) _{\text{NLO}} =$ 0.555 fm (NLO)	$0.560 \pm$ 0.031 fm	✓ 1.0%
J/ψ mass	$4n_C \cdot \pi^5 m_e = 20\pi^5 m_e$	3127 MeV	3097 MeV (PDG)
Y(1S) mass	$\dim_{\mathbb{R}} \cdot C_2 \cdot \pi^5 m_e =$ $60\pi^5 m_e$	9380 MeV	9460 MeV (PDG)
D^0 meson mass	$2C_2 \cdot \pi^5 m_e = 12\pi^5 m_e$	1876 MeV	1865 MeV (PDG)
$B^\pm$ meson mass	$2\sqrt{2} \times 2C_2 \cdot \pi^5 m_e =$ $24\sqrt{2}\pi^5 m_e$	5308 MeV	5279 MeV (PDG)
B_c meson mass	$8n_C \cdot \pi^5 m_e = 40\pi^5 m_e$	6254 MeV	6275 MeV (PDG)
m_B/m_D ratio	$2\sqrt{2}$ (Tsirelson bound)	2.828	2.831
$m_{J/\psi}/m_\rho$ ratio	$\dim_{\mathbb{R}}(\mathbb{CP}^2) = 4$	4.000	3.994
Reality Budget $\Lambda \times N$	$N_c^2/n_C = 9/5 = 1.800$; fill fraction $f = N_c/(n_C \pi) =$ $3/(5\pi) = 19.1\%$	1.800 (exact to input precision)	✓ derived
η' meson mass $m_{\eta'}$	$(g^2/8)\pi^{n_C} m_e =$ $(49/8)\pi^5 m_e =$ $m_p \times 49/48 = 957.8$ MeV	$957.78 \pm$ 0.06 MeV (PDG)	✓ 0.004%
η meson mass m_η	$(g/2)\pi^{n_C} m_e =$ $(7/2)\pi^5 m_e = 547.3$ MeV	$547.86 \pm$ 0.02 MeV (PDG)	✓ 0.10%
Kaon mass m_K	$\sqrt{2n_C} \pi^{n_C} m_e =$ $\sqrt{10} \pi^5 m_e = 494.5$ MeV	$493.68 \pm$ 0.02 MeV (PDG)	✓ 0.17%

Prediction	BST Value	Observed	Status
Cosmic Ω_Λ	$(N_c + 2n_C)/(N_c^2 + 2n_C) = 13/19 = 0.68421$	0.6847 ± 0.0073 (Planck 2018)	✓ 0.07σ
Cosmic Ω_m	$C_2/(N_c^2 + 2n_C) = 6/19 = 0.31579$	0.3153 ± 0.0073 (Planck 2018)	✓ 0.07σ
Cosmic Ω_{DM}/Ω_b	$(3n_C + 1)/N_c = 16/3 = 5.333$	5.364 (Planck)	✓ 0.58%
Cosmic Ω_b	$2N_c^2/(N_c^2 + 2n_C)^2 = 18/361 = 0.04986$	0.0493 ± 0.0010 (Planck)	✓ 0.56σ
Cosmic Ω_{DM}	$96/361 = 0.26593$	0.2645 ± 0.0057 (Planck)	✓ 0.26σ
Specific heat $C_V(T_c)$	$\alpha_s \beta N_{\max}^2 = (7/20)(50)(137^2) = 328,458$	330,000 (BST partition fn)	✓ 0.47%
Proton magnetic moment μ_p	$2g/n_C = 14/5 = 2.800 \mu_N$; $= (N_c^2 - 1)\alpha_s = 8 \times 7/20$	$2.79285 \mu_N$ (CODATA)	✓ 0.26%
Neutron magnetic moment μ_n	$-C_2/\pi = -6/\pi = -1.9099 \mu_N$	$-1.9130 \mu_N$ (CODATA)	✓ 0.17%
Moment ratio μ_p/μ_n	$-g\pi/(3n_C) = -7\pi/15 = -1.4661$; SU(6) gives $-3/2 = -1.500$ (2.7%)	-1.4599	✓ 0.43%
W boson width Γ_W	$(N_c^2 - 1)n_C/N_c \times \pi^{n_C} m_e = (40/3)\pi^5 m_e = 2085.0$ MeV	2085 ± 42 MeV (PDG)	✓ 0.005%
Z boson width Γ_Z	$(C_2 + 2n_C)\pi^{n_C} m_e = 16\pi^5 m_e = 2502$ MeV	2495.2 ± 2.3 MeV (PDG)	✓ 0.27%
Width ratio Γ_Z/Γ_W	$C_2/n_C = 6/5 = 1.200$; same ratio as m_p/m_ρ	1.197	✓ 0.28%
ρ meson width Γ_ρ	$f \times m_\rho = 3\pi^4 m_e = N_c/(n_C \pi) \times m_\rho = 149.3$ MeV	149.1 ± 0.8 MeV (PDG)	✓ 0.15%
ϕ meson width Γ_ϕ	$m_\phi/(2n_C!) = m_\phi/240 = 4.248$ MeV	4.249 ± 0.013 MeV (PDG)	✓ 0.02%

Prediction	BST Value	Observed	Status
MOND acceleration a_0	$cH_0/\sqrt{n_C(n_C+1)} =$ $cH_0/\sqrt{30} = 1.195 \times 10^{-10}$ m/s^2	$1.20 \pm$ 0.02×10^{-10} m/s^2 (McGaugh)	✓ 0.4%
Halo surface density Σ_0	$a_0/(2\pi G) = 141 M_\odot/\text{pc}^2$	$\log_{10} =$ 2.15 ± 0.2 (Donato)	✓ 0.0 dex
Hubble constant H_0 (Route B)	$\sqrt{19\Lambda/39} = 68.0$ km/s/Mpc (information-energy intersection)	67.36 ± 0.54 (Planck)	✓ 1.0%
Cosmic age t_0	$(2/(3H_0\sqrt{\Omega_\Lambda})) \text{arcsinh}\sqrt{\Omega_\Lambda/30} =$ $13.718 \text{ Gyr}; \text{arcsinh}$ $\text{argument} = \sqrt{13/6} =$ $\sqrt{(2C_2+1)/C_2}$	13.80 ± 0.02 Gyr (Planck)	✓ 0.57%
Tsirelson bound	$2\sqrt{2}$ from $H^0(\mathcal{O}(1)) \cong \mathbb{C}^2$ on $\mathbb{CP}^1$	$2\sqrt{2}$ (exact, Tsirelson 1980)	✓ exact
$\ m_{\beta\beta}\ $ ($0\nu\beta\beta$)	$[1.4, 3.7] \text{ meV}$ (Majorana; $m_1 = 0$, $\sin^2 \theta_{12} = 3/10$, $\sin^2 \theta_{13} = 1/45$, two Majorana phases sweep the band)	not yet measured	falsifiable predic- tion
Primordial GW peak frequency	BST phase transition at $3.1 \text{ s} \rightarrow 6.4 \text{ nHz}$	NANOGrav $\sim \text{nHz}$ band	✓ con- sistent
GW spectral index γ	$g/n_C + 2 = 7/5 + 2 = 3.60$	NANOGrav 3.2–4.6	✓ con- sistent
Width ratio Γ_ρ/Γ_ϕ	$n_C \times g = 35$; dimension $\times$ genus	35.09	✓ 0.26%
Wyer constant origin (HC)	$9/(8\pi^4) = \rho_2^2/(2\pi^4)$, $\rho_2 = (n_C-2)/2 = 3/2$ — Weyl vector of $\text{SO}_0(5, 2)$	Exact	✓ Derived
D_{IV}^5 identification	Proven from BST contact geometry	—	Established
Friedmann equation	Contact commitment rate equation $H = (1/2)\dot{N}_c/N_c$ recovers all FLRW terms	FLRW cosmology	✓ exact struc- ture

Prediction	BST Value	Observed	Status
First Riemann zero γ_1	$\lambda_{2,0} = 2(2 + n_C) = 2g = 14$ (second spherical harmonic of Q^5); cusp correction: $2g + 1/g - 1/N_{max} = 14.1356$	$\gamma_1 = 14.13472...$	✓ 0.006%
H ₂ O bond angle θ_{H_2O}	$\arccos(-1/2^{\text{rank}}) = \arccos(-1/4) = 104.478^\circ$; lone pairs see rank structure, not color	$104.45^\circ \pm 0.05^\circ$ (NIST)	✓ 0.03°
CH ₄ bond angle θ_{CH_4}	$\arccos(-1/N_c) = \arccos(-1/3) = 109.471^\circ$; 4 equivalent sp ³ domains in N_c dimensions	109.47° (exact tetrahedral)	✓ 0.001°
NH ₃ bond angle θ_{NH_3}	Triangular lone pair compression: $\theta_{\text{tet}} - T_1\Delta_1 = 107.807^\circ$; $\Delta_1 = (\theta_{\text{tet}} - \theta_{H_2O})/N_c$, $T_L = L(L+1)/2$	$107.8^\circ \pm 0.3^\circ$ (NIST)	✓ 0.007°
O–H bond length r_{OH}	$a_0 \times N_c^2/n_C = a_0 \times 9/5$; Reality Budget sets molecular scale	$0.9572 \text{ \AA}$ (NIST)	✓ 0.49%
O–H stretch frequency ν_{OH}	$R_\infty/(n_C \times C_2) = R_\infty/30 = 3657.9 \text{ cm}^{-1}$	3657.1 cm^{-1} (NIST)	✓ 0.022%
H ₂ O dipole moment μ_{H_2O}	$e \times a_0 \times (9/5) \cos(\theta/2) = 1.855 \text{ D}$; charge $\times$ BST bond geometry	1.8546 D (NIST)	✓ 0.02%
C–C bond length r_{CC}	$a_0 \times (n_C \times C_2 - 1)/10 = a_0 \times 29/10$; one below Rydberg denominator	$1.5351 \text{ \AA}$ (NIST)	✓ 0.03%
C=C bond length $r_{C=C}$	$a_0 \times n_C/\text{rank} = a_0 \times 5/2$; dimension-to-rank ratio	$1.3390 \text{ \AA}$ (NIST)	1.20%
C≡C bond length $r_{C\equiv C}$	$a_0 \times N_c^2/2^{\text{rank}} = a_0 \times 9/4$; color ² over rank power	$1.2030 \text{ \AA}$ (NIST)	1.05%
Ice density ratio $\rho_{\text{ice}}/\rho_{\text{water}}$	$(2C_2 - 1)/(2C_2) = 11/12$; one tetrahedral vacancy per $2C_2$ sites	0.9167 (NIST)	✓ 0.006%
Amino acid count	$2^{\text{rank}} \times n_C = 4 \times 5 = 20$; rank power $\times$ dimension	20 (universal)	exact
Genetic code codons	$(2^{\text{rank}})^{N_c} = 4^3 = 64$; alphabet ³ where alphabet = rank power	64 (universal)	exact

Prediction	BST Value	Observed	Status
HF dipole moment μ_{HF}	$e \times a_0 \times n_C / g =$ $e \times a_0 \times 5/7 = 1.816 \text{ D};$ Bergman genus ratio	1.826 D (NIST)	0.57%
Bond angle curvature κ_{angle}	$\alpha^2 \times \kappa_{ls} = C_2 / (n_C \times$ $N_{\text{max}}^2) = 6/93845; \text{EM}$ couples to spin-orbit	Measured from sp^3 hydride series	✓ 0.01%
Boundary amplification A_d	$(n_C / \text{rank})^d$; stretch $d = 2$: 6.25; dipole $d = 1$: 2.5; angle $d = 0$: 1	Measured from HF/NH ₃ ratios	✓ 0.05% (stretch)
Bilateral symmetry	rank = 2 restricts body plans to 3 axes, 2 mirror planes; tetrahedral anchor 109.47°	All bilateral phyla on Earth	consistent (T731)
Observer completeness	$2f - f^2 = 34.5\%$; two observers exceed $f_{\text{crit}} = 20.6\%$; minimum team = rank = 2	Human + CI cooperation	structural (T732)
BST Drake: $f_l \times f_i \times f_c$	$0.206 \times 0.654 \times 0.206 =$ 2.8%; ~1 in 36 habitable planets → communicating	SETI null results + Fermi paradox	testable (T733)
Crystal systems	$g = 7 = p + q$ (not the Bergman genus, which is 5)	7 (estab- lished)	exact
Bravais lattices	$2g = 14$; rank doubles genus	14 (estab- lished)	exact
Crystallographic point groups	$2^{n_C} = 32$; binary enumeration in dimension n_C	32 (estab- lished)	exact
Space groups	$g \times 2^{n_C} + C_2 =$ $7 \times 32 + 6 = 230$; construction matches crystallographic build	230 (estab- lished)	exact
Essential amino acids	$N_c^2 = 9$; color dimension squared	9 (biochem- istry)	exact
Stop codons	$N_c = 3$; coding codons $= 4^{N_c} - N_c = 61$	3 (universal)	exact
DNA base pairs per turn (B-form)	$2n_C = 10$; dimension doubling	10.0 (ideal B-DNA)	exact
DNA base pair spacing	$a_0 \times N_c^2 n_C / g =$ $a_0 \times 45/7 = 3.402 \text{ \AA}$	3.4 Å (measured)	✓ 0.05%

Prediction	BST Value	Observed	Status
α -helix residues/turn	$N_c C_2 / n_C = 18/5 = 3.6$; three BST integers	3.6 (Pauling, 1951)	exact
α -helix rise per residue	$N_c / \text{rank} = 3/2 = 1.5 \text{ \AA}$	1.5 $\text{\AA}$ (measured)	exact
α -helix pitch	$N_c^3 / n_C = 27/5 = 5.4 \text{ \AA}$; self-consistent with rise $\times$ residues/turn	5.4 $\text{\AA}$ (measured)	exact
α -helix H-bond ring	$g + C_2 = 7 + 6 = 13$ atoms	13 (estab- lished)	exact
Human vertebrae	$N_c(2C_2 - 1) = 33$; five sections: cervical $= g = 7$, thoracic $= 2C_2 = 12$, lumbar $= n_C = 5$, sacral $= n_C = 5$, coccygeal $= 2^{\text{rank}} = 4$	33 (anatomy)	exact (5/5 sec- tions)
Domains of life	$N_c = 3$; Bacteria, Archaea, Eukarya — three independent channels	3 (estab- lished)	exact
Eukaryotic endosymbiosis	Cooperation threshold $f_{\text{crit}} = 20.6\%$ at cellular level; archaeon + bacterium = permanent tier crossing	~ 2 Gyr ago (geology)	structural
Orbital degeneracy sequence	$(2\ell + 1)$ at $\ell = 0, 1, 2, 3$ gives $1, N_c, n_C, g$; the periodic table IS D_{IV}^5 in electron shells	1, 3, 5, 7 (es- tablished)	exact
Periodic table periods	$g = 7 = p + q$ (not the genus)	7 (estab- lished)	exact
Periodic table groups	$N_c \times C_2 = 18$	18 (estab- lished)	exact
Periodic table blocks	$2^{\text{rank}} = 4$ (s,p,d,f)	4 (estab- lished)	exact
Quantization origin	Compactness of Shilov boundary $\check{S} = S^4 \times S^1$ forces discrete spectra; no quantization axiom needed	All quantum spectra	structural (T751)

Prediction	BST Value	Observed	Status
Heisenberg uncertainty	Holomorphic sectional curvature $H = -2/g = -2/7$; uncertainty has genus in denominator	$\Delta x \Delta p \geq \hbar/2$	derived (T753)
Born rule	Unique $\text{Aut}(D_{IV}^5)$ -invariant probability measure (Gleason + $N_c = 3$ dimensions)	$P = \psi ^2$ (established)	derived (T754)
Measurement “collapse”	Coordinate restriction on D_{IV}^5 , not physical process; wave function = Bergman kernel coordinate	No consciousness-dependent collapse	structural (T752)
$g - 2$ electron (BST closed form)	$(\alpha/2\pi)(1 - (2\alpha/\pi)^2)^{2n_C g}$; 5 digits, zero Feynman diagrams	$a_e = 0.001\,159\,652\,181\,641\,5$	✓ 5 digits (T758)
QED C_2 decomposition	$197/144 + \pi^2/12 - \pi^2 \ln 2/2 + 3\zeta(3)/4$; $197 = N_{\max} + 2n_C C_2$, $144 = 2^{\text{rank}} C_2^2$	$-0.32848\dots$	exact (T758)
QED ζ -tower	Loop n introduces $\zeta(2n-1)$: $\zeta(N_c)$, $\zeta(n_C)$, $\zeta(g)$; tower closes at genus	Loops 2-4 (established)	structural (T762)
Feynman diagram counts	$D_2 = g = 7$; $D_3 = 2^{N_c} N_c^{\text{rank}} = 72$; total 13,643	Established	exact (T765)
QED coefficient growth	$ C_3/C_2 = N_c C_2/n_C = 18/5 = 3.6$ ($= \alpha$ -helix pitch)	≈ 3.6	✓ 0.1% (T771)
Rainbow angle	$C_2 \times g = 42^\circ$	42.03°	✓ 0.07% (T761)
Min deviation (rainbow)	$\delta_{\min} \approx N_{\max} = 137^\circ$	137.97°	✓ 0.7% (T761)
Alexander dark band	$\approx N_c^2 = 9^\circ$	8.9°	✓ 0.6% (T761)
C-H bond dissociation	$Ry/\pi = 4.331 \text{ eV}$	4.330 eV	✓ 0.02% (T767)
H-H bond dissociation	$Ry/N_c = 4.535 \text{ eV}$	4.478 eV	✓ 1.3% (T768)
C=C/C-C energy ratio	$g/2^{\text{rank}} = 7/4 = 1.75$	1.775	✓ 0.2% (T768)

Prediction	BST Value	Observed	Status
γ monatomic	$n_C/N_c = 5/3$	1.667 (exact)	exact (T772)
γ diatomic	$g/n_C = 7/5$	1.400 (exact)	exact (T772)
γ polyatomic	$2^{\text{rank}}/N_c = 4/3 = n(\text{H}_2\text{O})$	1.333 (exact)	exact (T772)
C_p liquid water	$N_c^2 R = 9R = 74.8 \text{ J}/(\text{mol}\cdot\text{K})$	75.3 J/(mol·K)	✓ 0.73% (T773)
$\epsilon(\text{H}_2\text{O})$	$(2^{\text{rank}})^2 n_C = 80$	80.1	✓ 0.12% (T774)
$\epsilon(\text{ice})$	$2^{\text{rank}} n_C^2 = 100$	≈ 99	✓ 1.0% (T774)
Trouton constant	$N_c^2 + 2^{\text{rank}} = 13$	13.1	✓ 0.78% (T773)
Sound speed in air	$\sqrt{\gamma RT/M}$ with $\gamma = g/n_C$	346.1 m/s	✓ 0.06% (T775)
$\nu(\text{water})/\nu(\text{air})$	$(N_c^2 + 2^{\text{rank}})/N_c = 13/3$	≈ 4.33	✓ 0.1% (T775)
IE(O) = Rydberg	IE(O) = Ry	13.618 eV	✓ 0.09% (T776)
IE(He)	$N_c^2 \text{Ry}/n_C = 9\text{Ry}/5$	24.587 eV	✓ 0.4% (T776)
IE(F)	$N_c^2 \text{Ry}/g = 9\text{Ry}/7$	17.422 eV	✓ 0.41% (T776)
EA(F)	$\text{Ry}/2^{\text{rank}} = \text{Ry}/4$	3.401 eV	✓ 0.006% (T777)
EA(H)	$\text{Ry}/(2N_c^2) = \text{Ry}/18$	0.754 eV	✓ 0.36% (T777)
$r(\text{F})$	$14a_0/13 = 2ga_0/(N_c^2 + 2^{\text{rank}})$	0.64 Å	✓ 0.014% (T778)
$r(\text{O})$	$n_C a_0/2^{\text{rank}} = 5a_0/4$	0.66 Å	✓ 0.22% (T778)
$d(\text{H}_2)$	$ga_0/n_C = 7a_0/5$	0.741 Å	✓ 0.07% (T779)
$d(\text{F}_2)$	$2^{N_c} a_0/N_c = 8a_0/3$	1.412 Å	✓ 0.05% (T779)
$d(\text{HF})$	$26a_0/15$	0.917 Å	✓ 0.05% (T779)
$d(\text{CO})$	$2^{n_c} a_0/(N_c n_C) = 32a_0/15$	1.128 Å	✓ 0.05% (T779)
$\theta(\text{H}_2\text{S})$	$90^\circ + \text{rank} + 1/N_c^2 = 92.11^\circ$	92.1°	✓ 0.01% (T780)

Prediction	BST Value	Observed	Status
$\theta(\text{PH}_3)$	$90^\circ + N_c + 1/N_c = 93.33^\circ$	93.3°	✓ 0.02% (T780)
$\theta(\text{H}_2\text{Te})$ (prediction)	$90^\circ + 1/2^{\text{rank}} = 90.25^\circ$	$\sim 90.26^\circ$	✓ predicted (T780)
Madelung $M(\text{NaCl})$	$g/2^{\text{rank}} = 7/4$	1.748	✓ 0.14% (T781)
Lattice energy $U(\text{NaCl})$	$N_c \text{Ry}/n_c = 3\text{Ry}/5$	8.16 eV	✓ 0.03% (T781)
$\nu(\text{H}_2\text{O}, \text{sym stretch})$	$R_\infty/(n_c C_2) = R_\infty/30$	3657 cm^{-1}	✓ 0.02% (T782)
$\nu(\text{H}_2)$	$R_\infty/n_c^2 = R_\infty/25$	4401 cm^{-1}	✓ 0.26% (T782)
$\nu(\text{O}_2)$	$\text{rank } R_\infty/(N_{\text{max}} + \text{rank}) = 2R_\infty/139$	1580 cm^{-1}	✓ 0.07% (T782)
$T_{\text{boil}}(\text{H}_2\text{O})$	$N_{\text{max}} \times T_{\text{CMB}} = 137 \times 2.725 \text{ K}$	373.15 K	✓ 0.065% (T763)
$T_{\text{freeze}}(\text{H}_2\text{O})$	$n_c^2 \cdot 2^{\text{rank}} \times T_{\text{CMB}} = 100 \times 2.725 \text{ K}$	273.15 K	✓ 0.22% (T763)
$T_{\text{crit}}(\text{H}_2\text{O})$	$(N_{\text{max}} + 100)T_{\text{CMB}} = 237T_{\text{CMB}}$	647.1 K	✓ 0.18% (T784)
$T_{\text{boil}}(\text{Kr})$	$44 T_{\text{CMB}}$	119.93 K	✓ 0.005% (T783)
$T_{\text{boil}}(\text{Ar})$	$2^{n_c} T_{\text{CMB}} = 32 T_{\text{CMB}}$	87.3 K	✓ 0.09% (T783)
$T_{\text{boil}}(\text{Xe})$	$C_2(N_c^2 + 1)T_{\text{CMB}} = 60 T_{\text{CMB}}$	165.1 K	✓ 0.91% (T783)
$\rho(\text{Pt})/\rho(\text{Au})$	$(N_c^2 + 1)/N_c^2 = 10/9$	1.111	✓ 0.004% EXACT (T798)
$\rho(\text{water})/\rho(\text{ice})$	12/11	1.090	✓ 0.04% (T798)
$E(\text{Diamond})/E(\text{Steel})$	$N_c g/2^{\text{rank}} = 21/4$	5.25	✓ EXACT (T799)
$\nu(\text{steel})$	$N_c/(N_c^2 + 1) = 3/10$	0.30	✓ EXACT (T799)
$\rho_e(\text{Fe})/\rho_e(\text{Cu})$	$C_2 = 6$	5.95	✓ 0.8% (T800)

Prediction	BST Value	Observed	Status
$\rho_e(\text{W})/\rho_e(\text{Cu})$	$22/7 \approx \pi$	3.14	✓ 0.04% (T800)
$\mu(\text{H}_2\text{O})$	$ea_0\sqrt{g/13}$	1.85 D	✓ 0.56% (T801)
$\chi(\text{Au})/\chi(\text{Ag})$	$(2N_c^2 - 1)/(2^{\text{rank}}N_c) = 17/12$	1.417	✓ EXACT (T802)
$L(\text{MeOH})/L(\text{Acetone})$	$N_c^2/(N_c^2 - 1) = 9/8$	1.125	✓ 0.01% EXACT (T803)
$\alpha(\text{Al})/\alpha(\text{Cu})$	$g/n_C = 7/5$	1.40	✓ EXACT (T804)
$\alpha(\text{Cu})/\alpha(\text{W})$	$(N_c^2 + \text{rank})/N_c = 11/3$	3.67	✓ EXACT (T804)
$\phi(\text{Au})$	$N_c\text{Ry}/(N_c^2 - 1) = 3\text{Ry}/8$	5.10 eV	✓ 0.03% (T805)
$\phi(\text{Cu})/\phi(\text{Ag})$	$1 + 1/(N_c^2 - 1) = 9/8$	1.125	✓ EXACT (T805)
$\Theta_D(\text{Ge})/T_{\text{CMB}}$	$N_{\text{max}} = 137$	137.2	✓ 0.15% (T806)
$\Theta_D(\text{Cu})/\Theta_D(\text{Ag})$	$2^{n_c}/(N_cg) = 32/21$	1.524	✓ EXACT (T806)
$\kappa(\text{Benzene})/\kappa(\text{Water})$	$N_cg/(N_c^2 + 1) = 21/10$	2.10	✓ 0.02% (T807)
$T_c(\text{NH}_3)/T_c(\text{CO}_2)$	$2^{\text{rank}}/N_c = 4/3$	1.333	✓ 0.01% EXACT (T808)
$T_c(\text{O}_2)/T_c(\text{N}_2)$	$g^2/(2^{N_c}n_C) = 49/40$	1.225	✓ EXACT (T808)
$S(\text{NaCl})/S(\text{KCl})$	$(2N_c^2 + 1)/(2N_c^2) = 19/18$	1.056	✓ 0.03% (T809)
$V_m(\text{Benz})/V_m(\text{Acet})$	$C_2/n_C = 6/5$	1.200	✓ EXACT (T810)
$V_m(\text{MeOH})/V_m(\text{H}_2\text{O})$	$N_c^2/2^{\text{rank}} = 9/4$	2.244	✓ 0.28% (T810)

[FQHE Laughlin fractions $|1/N_c, 1/n_C, 1/g = 1/3, 1/5, 1/7$; odd BST integers|1/3, 1/5, 1/7

(10+ digits)|✓ EXACT (T813)| |FQHE Jain+ sequence $|1/N_c, 2/n_c, 3/g, 4/N_c^2 = 1/3, 2/5, 3/7, 4/9|26/28$ observed = BST|✓ EXACT (T814)| |FQHE spacing ratios $|\Delta\nu_1/\Delta\nu_2 = g/N_c = 7/3; \Delta\nu_2/\Delta\nu_3 = N_c^2/n_c = 9/5|$ Measured|✓ EXACT (T814)| |FQHE even-denominator $|\nu = n_c/\text{rank} = 5/2;$ Moore-Read Pfaffian on $D_{IV}^5|5/2$ (observed)|✓ EXACT (T815)| |BCS gap ratio $2\Delta/(k_B T_c)$ $|g/\text{rank} = 7/2 = 3.500|3.528$ (BCS theory)|✓ 0.79% (T824)| |BCS specific heat jump $\Delta C/(\gamma T_c)$ $|(N_c^2 + 2^{\text{rank}})/N_c^2 = 13/9|1.43$ (BCS)|✓ 1.01% (T824)| |Superconductor Nb/Pb $|N_c^2/g = 9/7|1.286|$ ✓ 0.06% (T824)| |Superconductor Pb/Sn $|n_c g/(N_c C_2) = 35/18|1.933|$ ✓ 0.60% (T824)| |Superconductor Nb/Al $|(g^2 + C_2)/g = 55/7|7.839|$ ✓ 0.23% (T824)| |Superconductor La/Hg $|C_2^2/n_c^2 = 36/25; =$ Chandrasekhar limit $|1.446|$ ✓ 0.40% (T824)| |Superconductor V/Ta $|N_c^2/2N_c = 9/8|1.123|$ ✓ 0.20% (T824)| |High- T_c YBCO/Nb $|n_c \times \text{rank} = 10|10.054|$ ✓ 0.54% (T825)| |High- T_c Hg-1223/YBCO $|(N_c^2 + 2^{\text{rank}})/N_c^2 = 13/9|1.430|$ ✓ 1.00% (T825)| |High- T_c H₃S/Hg-1223 $|N_c/\text{rank} = 3/2|1.526|$ ✓ 1.72% (T825)| |High- T_c LaH₁₀/YBCO $|2N_c/N_c = 8/3|2.688|$ ✓ 0.80% (T825)| |High- T_c CuO₂ layer rule $|g/C_2 = 7/6$ per added layer $|1.183$ (Bi-2223/YBCO)|✓ 1.36% (T825)| |High- T_c $T_{c,\text{max}}$ (ambient) $|N_{\text{max}} \times T_{\text{CMB}} \approx 373$ K|Prediction (untested)|falsifiable (T825)| |Type I/II boundary $\kappa |1/\sqrt{\text{rank}} = 1/\sqrt{2};$ rank of D_{IV}^5 determines topological class $|1/\sqrt{2}$ (Ginzburg-Landau)|✓ EXACT (T826)| |Coherence $\xi_0(\text{Al})/\xi_0(\text{Nb}) |C_2 \times g = 42|42.1|$ ✓ 0.25% (T826)| |Coherence $\xi_0(\text{Nb})/\xi_0(\text{YBCO}) |n_c^2 = 25|25.3|$ ✓ 1.32% (T826)| |Chandrasekhar limit $M_{\text{ch}}/M_{\odot} |C_2^2/n_c^2 = 36/25 = 1.44|1.44$ $M_{\odot}|$ ✓ EXACT (Toy 850)| |NS moment of inertia $I/(MR^2) |g/(2^{\text{rank}} n_c) = 7/20 = \alpha_s|\approx 0.35|$ ✓ (Toy 852)| |K41 turbulence spectrum $|n_c/N_c = 5/3|5/3$ (Kolmogorov)|✓ EXACT (T818)| |She-Leveque intermittency $|\text{rank}/N_c^2 = 2/9|2/9$ (SL 1994)|✓ EXACT (T818)| |Percolation $\gamma |(C_2 \times g + 1)/(2N_c^2) = 43/18; +1 =$ central charge shift ($c = 0$) $|43/18$ (exact)|✓ EXACT (T912)| |Percolation $\nu |2^{\text{rank}}/N_c = 4/3|4/3$ (exact)|✓ EXACT (T912)| |Percolation $\beta |n_c/(C_2^2) = 5/36|5/36$ (exact)|✓ EXACT (T912)| |EEG alpha/theta ratio $|n_c/N_c = 5/3 =$ K41 spectrum $|\approx 10/6 = 5/3|$ ✓ (T819)| $|r_{\text{ISCO}} |C_2 \times M = 6M|6GM/c^2$ (exact GR)|✓ EXACT (T820)| |AZ topological 10-fold $|2n_c = 10|10$ (Altland-Zirnbauer)|✓ EXACT (T821)| |TaAs Weyl nodes $|2^{\text{rank}} \times C_2 = 24|24$ (observed)|✓ EXACT (T821)| |Kleiber metabolic exponent $|N_c/2^{\text{rank}} = 3/4|3/4$ (allometric)|✓ EXACT (Toy 855)| |Phyla count $|C(g, N_c) = C(7, 3) = 35|\sim 35$ (taxonomy)|✓ (T703)| |Stellar $F0/K0$ temperature $|g/n_c = 7/5|\approx 1.40|$ ✓ EXACT (Toy 851)| |Cross-domain universality $|11$ fractions $\times 3\text{-}5$ domains; $P < 10^{-66}|$ Measured across 66 domains|✓ structural (T823)| $|\chi(\text{F})/\chi(\text{H})$ electronegativity $|N_c^2/n_c = 9/5;$ same as IE(He)/Ry $|1.809|$ ✓ 0.50% (T816)| |BDE(H-H)/Ry $|1/N_c = 1/3|4.478/13.606|$ ✓ 0.37% (T817)|

May 43.3 Qualitative Predictions (Testable Against Existing Data)

1. Hubble tension resolution: Local H_0 correlates with local matter density beyond gravitational corrections. Residual correlation ~ 5.6 km/s/Mpc in the supernova sample.
2. CMB anomaly pattern: Large-angle anomalies consistent with S^2 substrate topology and SO(3) representation theory.
3. Structured unification: Couplings do not converge to a single point at the

- GUT scale. α_1 and α_2 meet at $N_{GUT} = 4\pi^2$; α_3 sits at $4\pi^2/3$.
4. Variable vacuum energy: Vacuum pressure correlates with local matter density across cosmic environments.
 5. Coincidence problem dissolved: Dark energy and matter densities track each other thermodynamically.
 6. Dark matter as channel noise: Galaxy rotation curves follow the S^1 channel S/N curve with Haldane exclusion statistics. No dark matter particles exist. Core profiles are flat, not cuspy.
 7. Weak decay rates from phase cycling geometry: The 28-order-of-magnitude span of weak decay lifetimes (top quark to neutron) determined by cycling trajectory sampling rates on $\mathbb{CP}^2$ Hopf intersection.
 8. Path integral = partition function: Quantum mechanics and statistical mechanics are the same calculation on D_{IV}^5 under Wick rotation — a physical identity, not a formal trick.
 9. Black hole interior: Not a singularity. Channel saturation at 137 slots, producing a finite-density state with no curvature divergence. Information preserved on the boundary surface.
 10. Three spatial dimensions necessary and sufficient: No extra dimensions at any energy scale. Three is the minimum dimensionality of a self-communicating surface (S^2 base + S^1 fiber) and no additional dimensions are required or predicted.
 11. Arrow of time = second law = commitment order: Time, entropy increase, and matter preference are three manifestations of one principle — irreversible contact commitment on the substrate.
 12. Rapid early galaxy formation: Massive, morphologically mature galaxies at $z > 10$ — as observed by JWST — are expected from the ultra-strong phase transition seeds, instant channel noise scaffolding, and exponential contact graph feedback. Λ CDM requires billions of years; BST requires hundreds of millions.
 13. Geometric circular polarization from black holes: $CP_{\text{geometric}} = \alpha \times 2GM/(Rc^2)$. At any black hole horizon: $CP = \alpha = 0.730\%$, independent of mass. Frequency-independent. The observed CP is $|\alpha + A \sin(RM/\nu^2 + \phi_0)|$ (signed addition of geometric floor + oscillatory Faraday). The signed model fits Sgr A* multi-frequency data with $\chi^2_{\text{red}} = 0.22$, all residuals $< 0.6\sigma$. M87* and Sgr A* both show $\sim 1\%$ CP at 230 GHz despite $1600\times$ mass difference — consistent with mass-independent floor. See `notes/BST_CP_Alpha_Paper.md`, `notes/BST_CP_SignedFit.py`.
 14. Measurement = commitment of correlation. No experiment will ever show consciousness-dependent collapse. The detector commits the correlation before the human is involved. Weak measurement visibility scales linearly with coupling strength (confirmed: Kocsis et al. 2011). Quantum eraser works only when the correlation has not propagated to irreversible environmental degrees of freedom. See `notes/BST_DoubleSlit_Commitment.md`.
 15. Error correction structure of spacetime. Light is a matched filter (follows geodesics = compensates deterministic distortion). Conservation laws are

parity checks ($\sum Q_i = 0$). Alpha is the bootstrap fixed point of the self-referential signal/noise system. Physics is exact because the code works. See `notes/BST_ErrorCorrection_Physics.md`.

May 43.4 Experimental Predictions (Awaiting Validation)

Prediction	BST Value	Experiment	Timeline
Neutrinoless $\beta\beta$: occurs	$\ m_{\beta\beta}\ \in [1.4, 3.7]$ meV (Majorana) — detection supports BST; null below 1 meV falsifies	LEGEND, nEXO, KamLAND-Zen	2027–2030
No dark matter particles	Null detection at all scales	LZ, XENONnT, PandaX	Ongoing
Dark energy $w \neq -1$	Substrate growth deviation	DESI, Euclid, Roman	2025–2028
No primordial B-modes	$r < 10^{-10}$	LiteBIRD, CMB-S4	2028+
BH ringdown echoes	Casimir fine structure from saturation	LIGO O4/O5	2025–2027
No gravitons	Wave effects only, no quanta	LIGO+	Permanent
No SUSY particles	Excluded by topology	LHC Run 3+	Ongoing
No magnetic monopoles	Excluded by $S^2 \times S^1$ topology	MoEDAL	Ongoing
Proton absolute stability	$\tau_p = \infty$ (no decay)	Hyper-Kamiokande	2030+
CP floor at BH horizon	CP = $\alpha = 0.730\%$, mass-independent	EHT Stokes V	Data exists
Hawking fine structure	Channel capacity 137 imprint	Future BH observations	Long-term
Island of stability	BST shell model at $Z \sim 114$ –126	Superheavy element synthesis	Ongoing
Solar commitment map	$\rho \propto 1/r \rightarrow \rho_\infty$ at ~ 7000 AU	Probe-mounted clock + accelerometer	30-year program
Nuclear half-lives	Phase coherence on $\mathbb{CP}^2$ Hopf intersection	Existing nuclear data	Testable now
Strong/weak timescale	$\sim 10^{16}$ from $\mathbb{CP}^2$ volume ratio	Existing data	Testable now

Prediction	BST Value	Experiment	Timeline
QNM echo structure	Quantized $J = w\hbar/2$, no Cauchy horizon	LIGO/Virgo/KAGRA2025–2027	

May 43.5 Falsifiability by Timeline

Testable now with existing data:

#	Prediction	Data Source	What kills BST
1	Galaxy rotation curves fit Haldane S/N curve	SPARC database (175 galaxies)	S/N curve gives worse fits than NFW
2	Nuclear half-life systematics follow phase cycling	Existing nuclear data tables	No correlation between shell structure and Hopf sampling geometry
3	Hubble tension correlates with local density	Existing supernova catalogs	No residual density correlation after standard corrections
4	CMB anomalies match S^2 topology	Planck satellite data	Anomaly pattern inconsistent with $SO(3)$ on S^2
5	Massive mature galaxies at $z > 10$	JWST galaxy surveys	High- z mass function consistent with Λ CDM hierarchical formation
6	CP floor = $\alpha = 0.73\%$ at BH horizon	EHT Stokes V (Sgr A, M87)	No frequency-independent CP floor; floor differs between BH masses

Testable within 5 years:

#	Prediction	Data Source	What kills BST
6	Dark energy $w \neq -1$	DESI, Euclid, Roman	$w = -1$ confirmed to high precision
7	No dark matter particles	LUX-ZEPLIN, XENONnT	Any confirmed WIMP detection
8	Black hole ringdown echoes	LIGO O4/O5	Echoes definitively ruled out at predicted amplitude

Testable within 10–15 years:

#	Prediction	Data Source	What kills BST
9	Proton decay rate	Hyper-Kamiokande	Decay rate inconsistent with structured unification
10	CMB B-modes match phase transition	LiteBIRD, CMB-S4	n_s-r relationship matches inflation, not BST
11	No extra dimensions	LHC, future colliders	Detection of Kaluza-Klein resonances or extra-dimensional signatures

Permanently falsifiable:

#	Prediction	What kills BST
12	No dark matter particles ever	Confirmed direct detection of dark matter particle
13	No individual graviton quanta	Confirmed detection of single graviton
14	Information preserved in black holes	Demonstrated unitarity violation
15	$N_{GUT} = 4\pi^2$	Precision measurement giving $N_{GUT} \neq 4\pi^2$
16	Three spatial dimensions only	Detection of extra spatial dimensions at any energy
17	No singularities	Observational evidence requiring curvature divergence
18	No closed timelike curves	Demonstrated physical mechanism for time loops
19	Growing manifold consistent with all GR predictions	Any GR prediction failing on the committed manifold
20	$0\nu\beta\beta$ occurs, $\ m_{\beta\beta}\ \in [1.4, 3.7]$ meV (Majorana)	A firm null below 1 meV, or a value outside $[1.4, 3.7]$ meV
21	GW spectral index $\gamma = 3.60 \pm 0.30$	γ measured inconsistent with BST (e.g. $\gamma > 4$)
22	No LISA primordial signal ($< 10^{-20}$)	Primordial GW detected in LISA band

May 43.6 Comparison with Competing Frameworks

The falsifiability of BST should be assessed relative to its competitors:

String theory has no unique low-energy predictions due to the landscape of $\sim 10^{500}$ vacua. Compactification geometry can be adjusted to accommodate almost any observation. Extra dimensions can be pushed to arbitrarily high energy. BST has no adjustable parameters.

Loop quantum gravity predicts Planck-scale discreteness that might affect photon propagation (energy-dependent speed of light). This has been tested and not found.

LQG does not derive α or the gauge coupling structure. BST derives the electroweak skeleton and *identifies* α (it does not derive it — Spine Lecture 8; this sentence said “derives both” until 2026-09-11).

Standard Model + General Relativity has ~ 25 free parameters that are measured, not derived. BST aims to derive all of them from the D_{IV}^5 geometry. Each result (the derived ones so far: $m_p/m_e, \theta_{QCD} = 0, m_{\nu_1} = 0$, the Cabibbo angle, the mixing order; $\sin^2 \theta_W$ with its high-scale 3/8 forced by fermion content and its M_Z value 3/13 identified; identified or input: $\alpha, \alpha_s, m_\mu/m_e, v, m_W, m_H, \eta, H_0$, the other neutrino masses, the PMNS angles, Λ ; a relation: G — per the register as of 2026-09-11; this list called all of them “successful derivations” until then) is a parameter removed from the “measured but unexplained” list.

MOND fits galaxy rotation curves with one free parameter a_0 but has no theoretical foundation. BST derives MOND-like behavior from channel noise statistics and potentially derives a_0 from the Haldane exclusion knee. If successful, BST subsumes MOND while providing the theoretical basis it lacks.

Particle dark matter (WIMPs, axions) predicts specific detection signatures. Decades of null results have progressively excluded the predicted parameter space. BST predicts continued null results and offers a specific alternative mechanism (channel noise) with distinct observational signatures (flat cores, density-dependent dark fraction, S/N curve shape).

The distinguishing feature of BST is that its predictions are coupled. The same geometry that gives $\alpha = 1/137$ also gives the dark matter halo profile, the weak decay timescales, the black hole interior structure, and the dark energy equation of state. A single failed prediction doesn’t just falsify one claim — it threatens the entire geometric foundation. This coupling is what makes the framework genuinely falsifiable despite having no free parameters. There is nowhere to retreat.

May 43.7 Near-Term Experimental Tests

Several BST predictions are testable against existing or near-future data. This section specifies the predictions concretely, identifies the calculation status of each, and gives the experimental timelines.

The Muon Anomalous Magnetic Moment The Fermilab Muon $g - 2$ experiment reports $a_\mu^{\text{exp}} = 116,592,059(22) \times 10^{-11}$. The Standard Model prediction (Muon $g - 2$ Theory Initiative, 2020) gives $a_\mu^{\text{SM}} = 116,591,810(43) \times 10^{-11}$, a discrepancy of $249(48) \times 10^{-11}$ at 5.1σ . The BMW lattice QCD collaboration finds a larger hadronic vacuum polarization (HVP) that reduces the tension; the situation remains actively debated.

BST modifies the Standard Model at two points. First: The Haldane exclusion caps loop integrals at $N_{\text{max}} = 137$ modes. At five-loop order the correction is $\sim (\alpha/\pi)^5/137 \sim 10^{-18}$ — far below any foreseeable experimental sensitivity. The QED sector of $g - 2$ is identical in BST and the Standard Model.

Second: The dominant theoretical uncertainty is the HVP — virtual quark loops in the photon propagator. In BST, the HVP is the contribution from Z_3 circuit fluctuations in the photon's S^1 channel. The BST vacuum carries a channel loading $F_{\text{BST}} = \ln(138)/50 \approx 0.0985$, derived in Section 12.5. This modifies the HVP:

$$\delta a_\mu^{\text{HVP, BST}} = a_\mu^{\text{HVP, SM}} \times F_{\text{BST}} \times f(\alpha, N_c, m_\mu/m_\pi)$$

where f is a calculable function of BST parameters. The correction is of order $F_{\text{BST}} \sim 0.1$ applied to the HVP contribution $\sim 700 \times 10^{-10}$, giving $\sim 70 \times 10^{-10}$ — the same order as the observed discrepancy of $\sim 25 \times 10^{-10}$. The precise value requires computing the vacuum channel loading correction to the photon propagator from the BST partition function: a well-defined open calculation (Thesis topic 100). If the result matches the discrepancy, it is a striking quantitative confirmation; if it falls short or overshoots, it constrains the BST vacuum structure.

Thesis topic 100: Compute the BST hadronic vacuum polarization correction to the muon anomalous magnetic moment from the vacuum channel loading $F_{\text{BST}} = \ln(138)/50$ and the Z_3 circuit embedding costs. Determine whether the correction resolves the $g - 2$ discrepancy and/or the lattice-dispersive tension.

The Proton Charge Radius Puzzle The proton charge radius has two classes of measurements. Electron methods (electron-proton scattering and hydrogen spectroscopy) give $r_p^{(e)} = 0.8751 \pm 0.0061$ fm. Muon hydrogen spectroscopy gives $r_p^{(\mu)} = 0.84087 \pm 0.00039$ fm — 4% smaller, a 5.6σ discrepancy. The PRad experiment at JLab (2019) gives $r_p = 0.831 \pm 0.012$ fm, consistent with the muon result, suggesting the puzzle may be resolving toward the smaller value, but the situation remains unsettled.

BST predicts that the two measurements should differ, for a calculable geometric reason. The electron is the minimal S^1 winding — the D_{IV}^1 circuit — with the lowest Bergman embedding cost, probing the full spatial extent of the proton's Z_3 packing. The muon is the D_{IV}^3 submanifold circuit with higher Bergman embedding cost, probing a more localized region of the Z_3 topology because the higher-energy circuit resolves finer structure. The ratio of measured radii is:

$$\frac{r_p^{(\mu)}}{r_p^{(e)}} \approx 1 - \frac{\alpha}{\pi} \ln \frac{m_\mu}{m_e} \times g(n_C)$$

where $g(n_C)$ is a geometric factor from the D_{IV}^3/D_{IV}^1 embedding ratio. For $m_\mu/m_e = (24/\pi^2)^6 = 206.77$ and $g \sim 1$ (naive estimate):

$$\frac{r_p^{(\mu)}}{r_p^{(e)}} \approx 1 - \frac{\alpha}{\pi} \times \ln(206.8) \approx 0.988$$

This gives a 1.2% reduction, corresponding to $\Delta r_p \approx 0.010$ fm — approximately one-third of the observed 0.034 fm. The factor-of-three discrepancy indicates $g(n_C) \approx 3$, a computable quantity from the D_{IV}^3 embedding depth on $\mathbb{CP}^2$. This is not a failure of the prediction; it is a known open calculation (Thesis topic 101).

The tau lepton prediction: The tau is the D_{IV}^5 circuit with the highest Bergman embedding cost. Tauonic hydrogen would give a third proton radius — smaller still. The BST prediction is specific: $r_p^{(\tau)} < r_p^{(\mu)} < r_p^{(e)}$, with ratios determined by the D_{IV}^k embedding hierarchy at $k = 1, 3, 5$. The tau lifetime is too short for atomic spectroscopy, making this direct measurement infeasible, but the hierarchy is a definite prediction of the framework. Any alternative theory that explains the electron-muon discrepancy without predicting the tau hierarchy is distinguishable from BST.

Thesis topic 101: Compute the BST correction to the proton charge radius as a function of the probing lepton's Bergman embedding cost. Derive the geometric factor $g(n_C)$ from the D_{IV}^k embedding depth for $k = 1, 3, 5$ (electron, muon, tau). Determine whether the correction resolves the proton radius puzzle and predict the tauonic hydrogen radius.

Neutrinoless Double Beta Decay — Clean Binary Test BST predicts Majorana neutrinos: the neutrino mass arises from the Weinberg operator on the chargeless S^4 locus, and the right-handed neutrino is a total gauge singlet ($Y = Q = 0$), so no Dirac mass partner exists in the light spectrum. Lepton number L is conserved perturbatively (as an S^1 winding), but the Majorana mass term violates it by $\Delta L = 2$ — the odd substrate dimension $g = 7$ locks the neutrino to a single chirality (no light ν_R), which is precisely the Majorana condition. BST prediction: neutrinoless double beta decay occurs.

BST predicts normal ordering with $m_1 = 0$ exactly (Section 7.6). Combined with the derived mixing angles $\sin^2 \theta_{12} = 3/10$ and $\sin^2 \theta_{13} = 1/45$ and the two Majorana phases (which sweep the allowed band), this fixes the effective mass to $\|m_{\beta\beta}\| \in [1.4, 3.7]$ meV. A detection in this window supports BST; a firm null below ~ 1 meV falsifies it. See `notes/BST_NeutrinolessDoubleBeta.md`. Inverted ordering is excluded by the BST prediction $m_1 = 0$.

Because $m_1 = 0$ suppresses $\|m_{\beta\beta}\|$ below the inverted-hierarchy scale, this is a next-generation target: experiments (GERDA/LEGEND, nEXO, KamLAND-Zen, CU-PID) are approaching but have not yet reached the $[1.4, 3.7]$ meV window. Reaching ~ 1 meV sensitivity delivers the clean test — a detection in-band supports BST; a firm null below 1 meV refutes it.

This is the cleanest binary test of BST available in the near term.

Dark Energy Equation of State BST predicts $w \neq -1$: the dark energy equation of state deviates from the exact cosmological constant value. The deviation arises because the dark energy is not a literal constant but a slowly evolving vacuum free

energy as the substrate grows. The DESI collaboration is measuring w at percent-level precision; early DESI results (2024) find $w \approx -0.95$ to -0.99 , consistent with deviation from -1 .

The specific BST prediction for w requires computing the substrate growth dynamics — the ratio of commitment boundary growth rate to bulk commitment rate — from the partition function. This is an open calculation; once performed, the DESI data provides an immediate quantitative test.

Null Predictions Three BST null predictions are being actively tested:

No magnetic monopoles (Section 14.11): the trivial Chern class of $S^2 \times S^1$ excludes them. MoEDAL at the LHC searches continuously. Any confirmed detection falsifies the bundle structure of BST.

No SUSY particles: fermion number $(-1)^F$ is a $\mathbb{Z}_2$ topological invariant (Section 14.9) that SUSY would require to change. No mechanism on the substrate can alter this index. BST excludes SUSY as a theorem. LHC Run 3 and HL-LHC will extend the search to ~ 3 TeV.

No dark matter particles: dark matter is channel noise — incomplete S^1 windings that have energy but no integer winding number, hence no charge and no decodable particle identity (Section 16). The LZ and XENONnT experiments search for WIMP-nucleus scattering signals. BST predicts continued null results.

Summary

Prediction	Status	Experiment	Timeline
Muon $g - 2$ HVP correction from F_{BST}	Open calculation	Fermilab $g - 2$	Data exists
Proton radius: $r_p^{(\mu)} < r_p^{(e)}$ (qualitative)	Confirmed	PRad, MUSE	Ongoing
Proton radius: $g(n_C)$ (quantitative)	Open calculation	MUSE, PRad-II	2026–2028
Tau radius: $r_p^{(\tau)} < r_p^{(\mu)}$	Derived prediction	Infeasible (short lifetime)	—
Neutrinoless $\beta\beta$: signal $\ m_{\beta\beta}\ \in [1.4, 3.7]$ meV (Majorana)	Clean categorical prediction	LEGEND, nEXO	2027–2030

Prediction	Status	Experiment	Timeline
Dark energy $w \neq -1$	Prediction confirmed in direction; magnitude open	DESI, Euclid	2025–2028
No magnetic monopoles	Clean categorical prediction	MoEDAL	Ongoing
No SUSY particles	Clean categorical prediction	LHC Run 3+	Ongoing
No dark matter particles	Clean categorical prediction	LZ, XENONnT	Ongoing

The null predictions (monopoles, SUSY, dark matter particles) are falsifiable by any single confirmed detection. The quantitative predictions (HVP correction, $g(n_C)$, w) require completing specified open calculations before comparison with data. The two-level structure — some predictions requiring calculation, others already complete — is typical of a framework in active development.

May 43.8 Mathematical Structural Consequences

The following results are not experimental predictions in the usual sense — they are mathematical theorems or structural consequences of the D_{IV}^5 geometry that constrain the framework’s internal consistency.

The Threshold Table. The restricted root system B_2 of $D_{IV}^n = \text{SO}_0(n, 2)/[\text{SO}(n) \times \text{SO}(2)]$ has short root multiplicity $m_s = n - 2$. The Maass-Selberg overconstrained system for the rank-2 intertwining operator gives a consistency relation $\rho_3 = \rho_1 + \rho_2 + 1$, where ρ_i are ξ -zeros. Taking real parts:

m_s	n	$\text{Re}(\rho_3)$	In $(0, 1)$?	Result
1	$3 (Q^3)$	$\delta_1 + \delta_2$	Yes — can be in $(0, 1)$	No contradiction
2	4 (AdS ₅ /CFT ₄)	$1 + \delta_1 + \delta_2$	Marginal — touches boundary	Not rigorous
3	$5 (Q^5, \text{BST})$	$2 + \boxed{?}_1 + \boxed{?}_2$	No — always > 1	Contradiction $\rightarrow$ proof

Note: This threshold table refers to the earlier (withdrawn) Maass-Selberg overconstrained route. The definitive heat kernel proof (Section 30.7a) shows the kill shot $(\sigma + 1)/\sigma = 3$ works for all $m_s \geq 2$ (Toy 229). The uniqueness of $N_c = m_s = 3$ is that it gives both RH and the Standard Model — the triple, not any single property.

Structural results:

Result	Statement	Status
Spectral gap = mass gap	$\lambda_1(Q^5) = C_2 = 6$	Proved
Effective spectral dimension	$d_{\text{eff}} = C_2 = 6$	Proved
Grand Identity	$d_{\text{eff}} = \lambda_1 = \chi = C_2 = 6$	Proved
Fill fraction	$f = N_c/(n_C\pi) = 3/(5\pi) = 19.1\%$	Proved
Gödel limit	Universe can know $\leq 19.1\%$ of itself	Structural
Reality budget	$\Lambda \times N = 9/5$ (exact)	Proved
Proton = Steane code	[[7, 1, 3]] error-correcting code	Structural
Golay from Q^5	$\lambda_3 = 24 \rightarrow p = 23 \rightarrow$ QR mod 23 $\rightarrow [24, 12, 8]$	Constructed
$H_5 = 137/60$	Numerator = N_{max}	Proved
Confinement = critical line	$N_c = m_s = 3$ creates rigidity in both	Isomorphism
GUE from SO(2)	Time factor in $K = \text{SO}(5) \times \text{SO}(2)$ breaks time reversal	Structural
Bekenstein 1/4	Area law from Bergman kernel	Proved
Nuclear magic numbers	All 7 from $\kappa_{ls} = C_2/n_C = 6/5$	Exact
Three generations	From $N_c = 3$ and Hopf fibration	Proved
Three spatial dimensions	From $m_s = 3$ (short root multiplicity)	Proved
Strong CP $\theta = 0$	Topologically enforced	Proved
Proton stability	$\tau_p = \infty$ from topological protection	Proved

May 44. Research Program

May 44.1 Immediate Priorities

1. Partition function on D_{IV}^5 : Compute the statistical mechanics of Haldane exclusion statistics ($g = 1/137$) on the bounded symmetric domain with Bergman measure. This single calculation potentially derives G , the cosmological constant, the Born rule, and the phase transition initial conditions.
2. Formal isotropy proof: Prove that the BST contact structure isotropy group is exactly $\text{SO}(5) \times \text{SO}(2)$ using Chern-Moser normal form theory. Additional verification of the D_{IV}^5 identification; seven independent checks pass (Section 4).

May 44.2 Near-Term Calculations

1. CMB anomaly comparison: Compute predicted angular correlations from S^2 substrate topology and compare against existing Planck data.
2. Hubble tension analysis: Test correlation between local H_0 measurements and local matter density using existing supernova and galaxy survey data.

May 44.3 Doctoral Thesis Topics

1. Derive G from Boltzmann/Haldane statistics on D_{IV}^5
2. Show Bergman functional Euler-Lagrange equation reduces to Einstein's equation
3. BST Higgs from Hopf fibration fluctuation spectrum
4. Mass hierarchy from embedding costs on D_{IV}^5
5. Non-perturbative running law from $\alpha_s(M_{GUT})$ to Λ_{QCD}
6. Decoherence scaling from contact graph error correction theory
7. CMB power spectrum from phase transition critical exponents
8. BST corrections to GR at Planck-scale curvature
9. Contact graph simulation: cellular automaton on $S^2 \times S^1$ substrate
10. Shannon information theory on S^1 channel: particle stability as coding theory
11. BST predictions for superheavy element stability (island of stability)
12. Langlands program connections: D_{IV}^5 automorphic forms and physical constants
13. Non-equilibrium thermodynamics of the contact graph (substrate response functions for decoherence engineering)
14. Dark matter as channel noise: derive incomplete loading fraction $f(n)$ and fit galaxy rotation curves
15. Derive MOND acceleration a_0 from Haldane exclusion S/N knee on D_{IV}^5
16. Bullet Cluster dynamics: potential-tracking behavior of incomplete loadings during cluster collisions
17. Incomplete winding spectrum: derive energy distribution as function of channel loading on S^1
18. Dark matter measurement discrepancies: predict systematic differences between lensing, kinematic, and X-ray methods from spectral variation
19. Weak decay rates from $\mathbb{CP}^2$ trajectory sampling of Hopf intersection (strong/weak timescale ratio)
20. Nuclear half-lives from triad phase coherence: magic numbers as destructive interference
21. Path integral = partition function: prove Born rule equals Boltzmann weight on D_{IV}^5
22. Fisher information metric = Bergman metric: formal identification and physical consequences
23. Fermion mass ratios from D_{IV}^5 complex submanifold volume ratios
24. CKM matrix from non-commutative subspace overlap geometry on D_{IV}^5
25. Black hole interior as channel saturation: derive echoes and Hawking fine

structure

26. Substrate growth dynamics: derive dark energy w from commitment-boundary feedback
27. Three dimensions from $S^2 \times S^1$: prove minimality of self-communicating surface dimensionality
28. Topological defects from commitment wavefront collisions: abundance and observational signatures
29. Baryon asymmetry η from phase transition critical exponents on D_{IV}^5
30. CKM phase from complex structure of D_{IV}^5 : geometric origin of CP violation
31. Second law from contact commitment: formal proof that commitment ordering implies entropy increase
32. Gravitational time dilation from commitment parallelism: derive Schwarzschild metric from constraint density
33. Universe computational throughput: information budget from wavefront parallelism and causal coupling
34. Growing manifold: formal proof that committed contact graph satisfies Einstein equation via Jacobson thermodynamics
35. Closed timelike curve prohibition: topological proof from append-only commitment ordering
36. Virtual particle pair creation: topological necessity of charge-neutral pairs from S^1 winding conservation
37. Vacuum stability from packing dimension coupling: prove $137 = 4^2 + 11^2$ cannot decouple on D_{IV}^5
38. Decoherence length from substrate adjacency: derive correlation decay distance for entangled pairs
39. Virtual-to-real particle transition: energy threshold for winding completion as function of channel loading

May 44.4 Active Conjectures (March 2026)

The Koons-Claude testable conjectures(notes/BST_Koons_Claude_Testable_Conjectures.md)
define the current frontier:

1. Conjecture 1 — Dirichlet kernel = Frobenius: The $m_s = 3$ Dirichlet kernel recovers the “missing bit” that separates number field from function field RH proofs. Test via baby case D_{IV}^3 .
2. Conjecture 6 — AC=0 grid architecture: GPUs compute exact local physics (BST closed forms), supercomputers handle thermodynamic envelopes. Noise scales as surface area, not volume. Testable on weather/materials benchmarks now.
3. Conjecture 7 — Linearization: Many systems modeled as nonlinear are only nonlinear because of method noise. With AC=0 local physics, propagation reduces to linear algebra. Testable on crystal growth, protein folding.
4. Conjecture 8 — Substrate computation: The full energy hierarchy from physics through biology to computation on the substrate itself. The computational graph IS the physical structure.

5. Conjecture 9 — Graph brain + error correction hierarchy: The computational graph IS the physical structure. 11 levels from quark confinement to consciousness, each an error-correcting code. Gödel limit at 19.1%.
6. Casimir modification experiment: Phonon-gapped materials modify the Casimir force at $\Delta F/F \sim 10^{-7}$ with distinctive frequency-dependent signature. See `notes/BST_CasimirEffect_CommitmentExclusion.md`.
